163 Focal Therapy of Prostate Cancer

Kae Jack Tay, MBBS, MRCS, MMed, MCI, FAMS, Sriram Deivasigamani, MBBS, MS, MRCS, and Thomas J. Polascik, MD, FACS

The global incidence of prostate cancer is estimated at 30.6 per 100,000 per year, and it is the second-most-incident cancer among males (Ferlay et al., 2013). In the United States, it is the most common solid organ malignancy among males and the third leading cause of death (American Cancer Society, 2016). In a similar pattern, the incidence of prostate cancer has been observed to increase over a 10-year period, particularly in countries with a high human development index and per capita gross domestic product (Wong et al., 2016). Much of the controversy that has arisen regarding prostate cancer overdiagnosis and overtreatment is due to the concern that many more patients have been exposed to the side effects of traditional radical treatments than necessary. Although these treatments have been highly effective in the oncologic sense, there is often collateral damage to the surrounding cavernosal nerves, urinary sphincter, and other adjacent organs (Resnick et al., 2013). Tools for early detection of prostate cancer, such as multiparametric magnetic resonance imaging (mpMRI) and MRI-fusion targeted biopsies, as well as the recognition that Gleason grade 4 and 5 cancers are the biologically lethal subgroup of cancers, have now led to an interest in the use of gland-sparing targeted therapies to achieve oncologic control with less morbidity (Mendez et al., 2015).

Focal therapy (FT) refers to the guided ablation of a cancerous lesion that is image defined and biopsy confirmed, or only biopsy confirmed, with a safety margin surrounding the target lesion.

Concepts and Historical Perspective

The idea of prostate FT followed the heels of the successful paradigm shift in breast cancer treatment among breast surgeons. The traditional Halsted mastectomy had given way to breast preservation utilizing local excision and reconstructive techniques. Early attempts in prostate gland–sparing treatments involved subtotal ablations, whereby the sparing of one, or possibly both, neurovascular bundles was attempted (Onik et al., 2002). The term *male lumpectomy* was then coined to describe hemiablation of the prostate as the first form of FT (Mouraviev and Polascik, 2012; Onik et al., 2007a). Many early ablations were performed without the benefit of advanced imaging, and lesion localization techniques varied from standard transrectal biopsies to 5-mm transperineal template grids. Steady advances in technology now allow surgeons to better visualize the prostate and localize the cancer, making possible true image-guided focal destruction of the offending cancer with a margin of normal tissue.

Treatment of prostate cancer is contingent on a balance between cancer aggressiveness and life expectancy. The goal of FT is to treat only the foci of cancer within the prostate gland that will affect the patient's survival or quality of life while preserving surrounding tissue and structures and, in turn, the patient's sexual and urinary function.

Biologic Basis for Prostate Focal Therapy

Using Cancer Grade as an Indication of Aggressiveness

Natural history studies suggest that tumor grade, serum prostate-specific antigen (PSA) levels, and clinical stage are the best predictors of prostate cancer–related mortality. Among these phenotypic expressions of aggressivity, cancer grade remains the earliest indication of lethal potential in low-volume, localized prostate cancer. A modification of the Gleason grading system described by Donald Gleason in 1973 remains the most effective and practical prognosticator of cancer aggressiveness today.

Hanahan and Weinberg (2000) defined the hallmarks of cancer in a 2000 review paper, further expanded in 2011 (Hanahan and Weinberg, 2011), listing the acquisition of cell self-sustenance, immortality, and invading capabilities as the distinguishing features of cancer (Fig. 163.1). Gleason grade 4 and 5, compared to grade 3 cancers, do possess many characteristic molecular hallmarks of malignancy.

Unopposed growth is a key feature in high-grade prostate cancers. Ross et al. (2011) described 670 genes expressed differentially between Gleason 3+3 and Gleason 4+4 cancers isolated by laser capture microdissection. These included genes involved in the developmental processes, signal transduction, chemokine, and embryonal stem cell pathways, particularly involving the androgen receptor, epidermal growth factor receptor (EGFR), tumor necrosis factor-α (TNF-α), and Notch signaling cascades. This explained earlier findings (Skacel et al., 2001) that EGFR gene copy numbers were not increased when examined using fluorescent in situ hybridization (FISH) of Gleason pattern 3 and 4 areas obtained from Gleason 3+4 radical prostatectomy specimens. However, the same investigators did find that aneusomy of chromosomes 7, 8, and 17 and amplification of HER-2/neu occurred almost exclusively in pattern 4 areas, further supporting the view that Gleason grade 4 carries a higher malignant potential compared to grade 3.

Loss of tumor suppression is an important cause of unopposed growth. Most gene deletions in prostate cancer typically involve large chromosomal segments. Kluth et al. demonstrated that the size of heterozygous deletions of 6q and 16q, the most commonly deleted segments in prostate cancer and those that contain large numbers of tumor suppressors, were highly correlated with Gleason score (Kluth et al., 2017).

Loss of cell cycle regulation is also a prominent feature of cancers with a higher Gleason grade. Cyclin D2, a critical cell cycle regulatory gene (Susaki and Nakayama, 2007), has also been found to be inactivated via promotor methylation at greater levels where Gleason score was higher (Padar et al., 2003). Loss of cyclin-dependent kinase inhibitor p27(kip1) protein is also correlated with Gleason grade (Guo et al., 1997).

Higher-Gleason-grade cancers are more likely to exhibit anti-apoptotic features compared to their low-grade counterparts. True et al. (2006) discovered an 86-gene panel differentiating Gleason pattern 3 from Gleason pattern 4 or 5 prostate cancer cells obtained from laser capture microdissection of radical prostatectomy specimens. Among this panel, they reported a specific finding of more intense staining in grade 4 and 5 cancers for the defender against apoptosis 1 *(DAD1)* gene, a downstream target of the nuclear factor-κB (NFκB) survival pathway that exhibits antiapoptotic functions (Patel et al., 2000) and has been associated with perineural invasion–associated prostate cancer in vitro (Ayala et al., 2004). Fleischmann et al. (2012) showed that Bcl-2, an antiapoptotic gene known to have increased gene copy numbers in androgen-independent prostate cancer, was already overexpressed in the hormone-naive setting and especially so in cancers with high Gleason score, advanced stage, and high proliferation index.

Aggressive cancers have also been found to have increased angiogenesis, with vessels that are primitive in morphology and function. Mucci et al. (2009), using immunostaining of CD34 to identify and

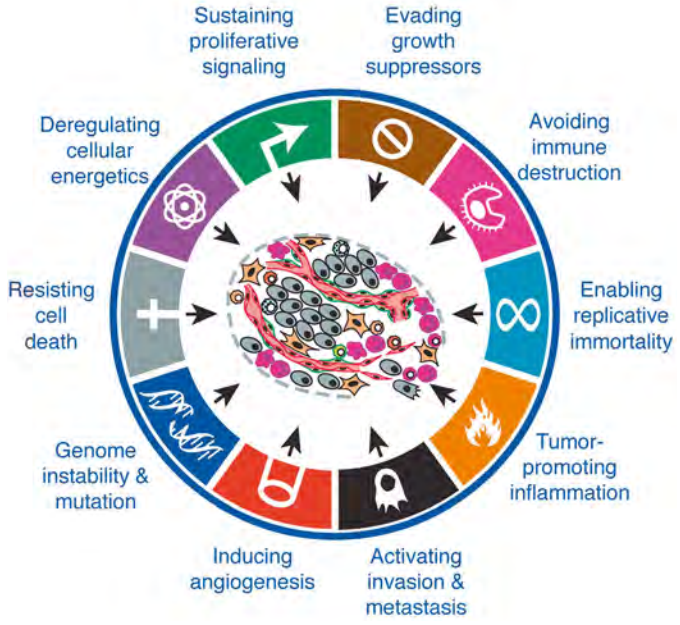


Fig. 163.1. Emerging and established hallmark capabilities and enabling characteristics of cancer. (Modified from Hanahan D, Weinberg RA: Hallmarks of cancer: the next generation, *Cell* 144(5):646–674, 2011.)

measure vessel density in 572 prostatectomy tumor tissue blocks, reported that males with tumors exhibiting the smallest vessel diameter and the most irregularly shaped vessels were 5 and 17 times more likely to die from lethal cancer, respectively. Angiogenic growth factors, such as vascular endothelial growth factor (VEGF)/protein serine kinase 1 (PK-1) expression, are also progressively expressed in prostate cancer as Gleason score increases (Pasquali et al., 2006). Angiogenesis, through increased sustenance and providing a path for metastasis, contributes to the lethal potential of high-Gleason-grade cancers.

Metastasis is a hallmark of malignancy, and epithelial-mesenchymal transition (EMT) is thought to be an important step in the development of pathological migratory properties. Jedroszka et al. (2017) found that the expression profile of 43 genetic EMT markers was strongly associated with Gleason score. Secreted frizzled-related protein 4 (SRFP-4), a regulator of the Wnt signaling pathway that is implicated in carcinogenesis as well as EMT (Sandsmark et al., 2017a), was found to be significantly more expressed in high-Gleason-grade groups (3–5) and in lethal and metastatic prostate cancers (Sandsmark et al., 2017b).

In summary, Gleason patterns 4 and 5 demonstrate many of the hallmarks of malignancy compared to Gleason pattern 3, and these represent priority targets for elimination. In early-stage, low-volume prostate cancer, focal treatment of these areas may be able to eradicate potentially lethal cancer, with the remainder of the benign/pattern 3–bearing gland being subjected to surveillance.

Multifocality of Prostate Cancer Versus the Index Lesion Hypothesis

Upon pathological examination of radical prostatectomy specimens, it appears that the majority of prostate cancer (50-91%) is multifocal, except for one Korean study (Song et al., 2003) reporting a unifocal cancer rate of 67% (Table 163.1). However, it is thought that the IL, the largest or highest-grade tumor focus within the prostate, determines the subsequent course of the cancer (Ahmed, 2009). This hypothesis is supported by the following data:

1. Most of the time, the grade/ stage is determined by the IL.
   In a series of 961 radical prostatectomy specimens, Ohori et al. (2006) reported that 377 lesions exhibited extracapsular extension, of which 84% were from the main or IL, whereas 16% were from nonindex lesions. Analyzing the subpopulation of 470 males with a serum PSA <10 ng/mL, they found that 92% of the time, extracapsular extension occurred in the IL only, 3% in the nonindex lesions, and 5% in both.
2. Most of the tumor volume is contributed by the IL.
   Ohori et al. (2006), reporting on 961 radical prostatectomy specimens, found that the IL accounted for 80% of the tumor burden. In a series of 61 radical prostatectomies, Hollmann et al. (2015) reported that the IL accounted for 88% of total tumor volume within the prostate. Both total and index tumor volumes have been found to be important predictors of biochemical recurrence after radical prostatectomy (Shin et al., 2016), and the data suggest that most of the intraglandular tumor volume can be dealt with by treating the IL.
3. Majority of satellite tumors are small and low grade.
   The definition of an insignificant cancer using <0.5 mL as a cutoff was created by Stamey et al. in 1993—a conclusion based on findings that (a) Surveillance, Epidemiology, and End Results (SEER) data suggested the lifetime probability of significant prostate cancer to be 8%, and (b) the largest 8% of cancers in the Stanford University bladder cancer cystoprostatectomy series were of the volume 0.5 to 6.1 mL. Among six radical prostatectomy series (Catto et al., 2011; Hollmann et al., 2015; Karavitakis et al., 2011, 2012; Noguchi et al., 2003; Wise et al., 2002) where the size of the satellite lesions was measured, the incidence of unifocal cancer ranged from 12.4-25%, but when accounting for cases where satellite lesions were <0.5 mL in aggregate, this figure rose to between 52% and 93.4%. Villers et al. (1992) found that 80% of secondary cancers were less than 0.5 mL in volume. Application of Stamey's rationale to modern series such as the Charing Cross cysto-prostatectomy series or the Rotterdam arm of the European Randomized Screening for Prostate Cancer trial yields thresholds of >1.09 mL (Winkler et al., 2007) or >1.31 mL (Wolters et al., 2011), respectively, for clinically significant cancer. This suggests that if these small lesions of doubtful clinical significance were to be left untreated and on observation, a much larger proportion of patients would be eligible for FT than the earlier pathological data suggest.
4. Genetic studies suggest a monoclonal origin of metastatic or lethal prostate cancer.
   Besides grade heterogeneity among the multiple lesions in multifocal prostate cancer, these lesions tend to be diverse in gene mutations (Cooper et al., 2015), copy number (Ibeawuchi et al., 2013), methylation (Serenaite et al., 2015), and genetic expression (Wei et al., 2017), suggesting the presence of multiple cancer lineages within the prostate that may have differing aggressivity. Further, Liu et al. (2009), analyzing prostate cancer metastasis in autopsies of patients who had died from prostate cancer, found that most, if not all, metastatic prostate cancers have a monoclonal origin and maintain the unique signature copy number pattern of the parent cancer cell while also accumulating separate subclinical, sustained changes (Fig. 163.2).

In summary, if the IL can be accurately identified and ablated, the majority of tumor within the prostate and its most aggressive components can be treated. By removing the lethal component of the prostate cancer, theoretically, metastasis and death could be prevented.

On the other hand, it is important to note that pathological features of aggression, such as largest tumor size, highest Gleason score, or highest stage, do not always occur in the same nodules and in fact may occur in satellite, rather than index, lesions 12-38% of the time. Ruitjer et al. (1996) found that 18% of ILs had a Gleason score lower than that of adjacent nonindex lesions, and extraprostatic extension (EPE) was found in two nonindex lesions. Arora et al. (2004) found that the Gleason score of the index lesion was correlated with the overall Gleason score in only 68% of specimens. Huang et al. (2014) found that in most of the multifocal cases (156/176, or 88.7%), the highest Gleason score, tumor volume, and presence of EPE were found in the index lesion. To further

TABLE 163.1 Whole-Mount Histology Studies Demonstrating the Multifocality of Prostate Cancer

AUTHOR	POPULATION/ INSTITUTION	YEAR PUBLISHED	PERIOD UNDER REVIEW	*N*	PSA	% PT2	% UNIFOCAL	% UNIFOCAL AND PT2	% UNIFOCAL WITH SATELLITE TUMORS < 0.5 ML	% MULTI-FOCAL
Villers et al. (1992)	Stanford University, USA	1992	1983–1989	234	—	—	50	—	—	50
Miller et al. (1994)	University of Colorado, USA	1994	—	151	—		43.7	—	—	56.3
Ruijter et al. (1996)	University Hospital Nijmegen, the Netherlands	1996	1992–1994	61	—	78	28	53	—	72
Djavan et al. (1999)	University of Vienna, Austria	1999	—	206	—	—	33.1	—	—	66.9
Chen et al. (2000)	MD Anderson Cancer Center, USA	2000	1990–1996	180	Mean 9.9 median 7.5	—	17	—	—	83
Rukstalis et al. (2002)	MCP Hahnemann University School of Medicine, Philadelphia, Pennsylvania, USA	2002	1998–2000	112	Median 7 (0.7–200)	—	20.5	—	—	79.5
Wise et al. (2002)	Stanford University, USA	2002	1992–1996	486	Mean 3.37 (2.2)	—	17	—	75	83
Noguchi et al. (2003)	Stanford University, USA (nonpalpable cancer subgroup)	2003	1988–1997	222	Mean 8.8 (4–50.4)		24		63	76
Song et al. (2003)	Samsung Medical Center, Korea	2003	1995–2002	132	Mean 16.4 (24.7)	50	66.7			33.3
Ng et al. (2004)	Indiana University School of Medicine, USA	2005	1999–2003	364	Median 6.2 (mean 8.4, 0.3–150)	68	14.8	—	—	85.2

Continued

TABLE 163.1 Whole-Mount Histology Studies Demonstrating the Multifocality of Prostate Cancer—cont'd

AUTHOR	POPULATION/ INSTITUTION	YEAR PUBLISHED	PERIOD UNDER REVIEW	*N*	PSA	% PT2	% UNIFOCAL	% UNIFOCAL AND PT2	% UNIFOCAL WITH SATELLITE TUMORS < 0.5 ML	% MULTI-FOCAL
Horninger et al. (2004)	University of Innsbruck, Austria (Tyrol PSA screening project)	2004	—	82	Mean 3.1	—	35	—	—	65
Iczkowski et al. (2008)	University of Colorado, USA	2008	2000–2003	393	Median 5.5 (2.5–48.7)	89	23	—	—	77
Rice et al. (2009)	Walter Reed Army Medical Center, USA	2009	1993–2008	1056	Mean 7.6	58	8.9	51	—	99.1
Karavitakis et al. (2011)	Imperial College, UK	2011	2001–2003	100	Mean 9.5 (1.1–28)	76	22	68.2	87.1	78
Catto et al. (2011)	ProTect study participants, UK	2011	—	525	Mean 6.1 (3.6)	68.7	19	86.4	33	81
Karavitakis et al. (2012)	Imperial College, UK	2012	2007–2009	95	Mean 11.9	62	25	67.4	52	75
Huang et al. (2014)	New York University Langone Medical Center, USA	2014	2011–2012	201	Mean 8.3 (0.5–19)	62.2	12.4	72	—	87.6
Le et al. (2015a)	University of California, Los Angeles, USA	2015	2010–2013	122	Mean 6.0 (4.8–8.3)	71	36	—	—	64
Hollmann et al. (2015)	The Netherlands Cancer Institute, the Netherlands	2015	2010–2012	61	Median 11 (4.1–37)	69	16	70	93.4	84

PSA, Prostate-specific antigen; *pT2,* pathological tumor stage 2.

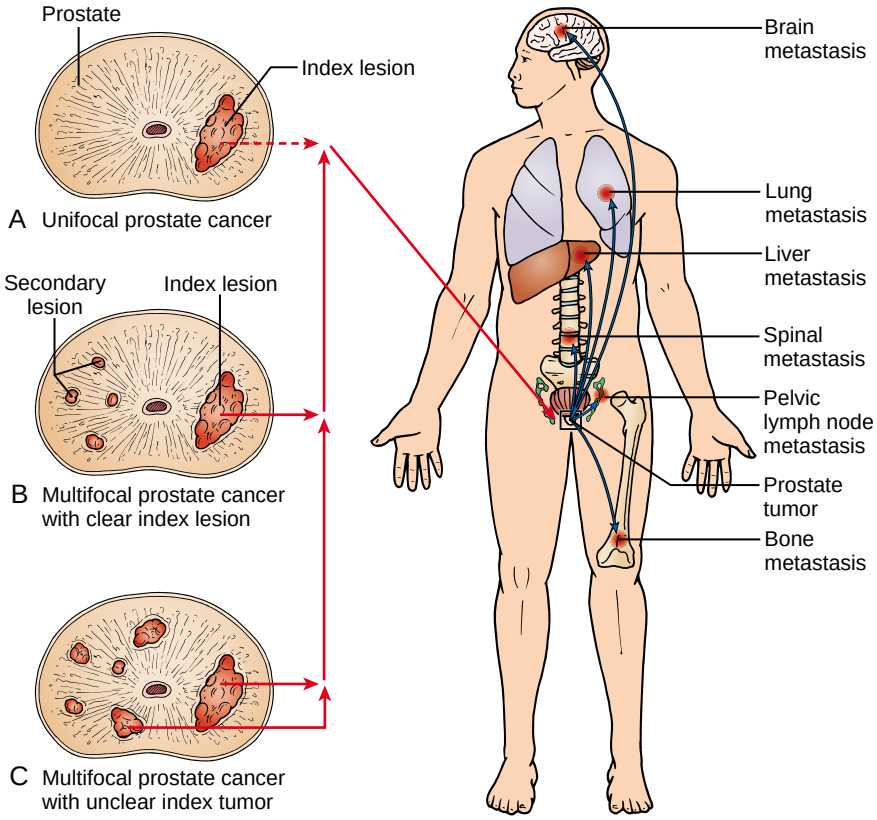


Fig. 163.2. (A) Unifocal prostate cancer. (B) Multifocal prostate cancer with clear index lesion and one or more separate secondary tumor foci with smaller volumes (most common). (C) Multifocal cancer with unclear index tumor. (From Perera, Krishnananthan N, Lindner U, Lawrentschuk N: An update on focal therapy for prostate cancer, *Nat Rev Urol* 13(11):641–653, 2016.)

confound matters, in one case report of a patient dying from metastatic prostate cancer, the lethal clone was traced back to a small, low-grade (Gleason pattern 3) focus rather than other high-grade lesions (Haffner et al., 2013).

These biologic variations may contribute to treatment failures in FT applied to the index lesion and are a reminder that close post-treatment surveillance is necessary so that salvage interventions can be undertaken in a timely manner. These same biologic exceptions can also be associated with treatment failure with conventional whole-gland approaches and active surveillance (AS). Nonetheless, the advances in biologic understanding have driven contemporary clinical concepts in FT, which will undoubtedly continue to evolve as we further our knowledge in this area.

CLINICAL APPLICATIONS OF FOCAL THERAPY

Treating the Index Lesion

The concept that the natural history of prostate cancer is typically driven by a primary lesion rather than secondary lesions (Stamey et al., 2001) is being increasingly accepted. If the dangerous foci in the prostate could be eradicated, oncologic control of the cancer could be achieved while better maintaining quality of life. The secondary lesions in the remainder of the prostate gland could then be placed on AS.

The Complementary Roles of Focal Therapy and Active Surveillance

Two of the largest active-surveillance cohorts have reached 10 to 15 years of follow-up, showing an extremely low metastasis rate among patients with D'Amico low-risk prostate cancer (Klotz et al., 2015; Tosoian et al., 2015). In one of the cohorts with more inclusive criteria (up to low-volume Gleason 3+4 cancer), it was found that the majority of those in whom cancer progressed had Gleason 3+4 disease (Yamamoto et al., 2016). In the Prostate Intervention Versus Observation Trial (PIVOT), a survival benefit was seen in the intermediate-risk subgroup with prostatectomy rather than observation (Wilt et al., 2012). Indeed, Gleason grade 4 cancers appear to have many more hallmarks of cancer compared to Gleason grade 3 and likely represent the relevant therapeutic target (Klotz, 2013).

These findings support a complementary strategy of treating intermediate- or high-grade cancer foci with FT while monitoring the remainder of the gland having low-grade cancer with AS (Klotz and Polascik, 2014). During surveillance, if any higher-grade cancers are detected, they could potentially be treated focally again. This would provide long-term oncologic control and effectively transform prostate cancer into a chronic disease, commensurate with its long natural history.

To these ends, a successful focal strategy depends on a multidisciplinary team of uro-oncologists, radiologists, pathologists, and others to do the following:

1. Accurately determine the disease extent/location through advanced imaging and biopsy.
2. Ascertain that the patient will benefit from treatment and be compliant with follow-up.
3. Completely ablate of the IL(s).
4. Monitor the patient after treatment utilizing advanced imaging and biopsy, with a view to future targeted treatment of either persistent or de novo disease or conversion to whole-gland treatment as necessary.

PATIENT SELECTION

Patient selection is the fundamental tenet of achieving a successful FT strategy. The aggressive lesion within the prostate must be identified

and its extent carefully delineated to completely ablate it. Because of the higher risk of cancer recurrence/persistence compared to radical therapy, the patient must be compliant, willing, and psychologically suited to being closely surveilled after treatment.

The lesion is first identified with a combination of histologic and/or radiologic methods. Advanced radiologic techniques such as mpMRI allow the entire prostate gland to be visualized and, in some cases, predict a more aggressive histology. However, because of limitations in current technology, there remains a consensus to histologically confirm the aggressiveness of the lesion through biopsy. Furthermore, current imaging studies remain imperfect in predicting the absence of aggressive cancer, and many experts would advocate a systematic biopsy to evaluate the gland more comprehensively.

Biopsy-Based Lesion Identification

Much of the work in biopsy-based lesion localization was directed toward intraglandular cancer lateralization for the purposes of focal hemiablation. At this time in the evolution of FT, focal hemiablation was reserved primarily for low-grade cancers, many of which might be placed on AS today. However, the data provide valuable insight on the ability of biopsy to accurately map cancer. Among patients who had undergone radical prostatectomy, Polascik et al. (2009) showed that transrectal ultrasonography (TRUS)-guided biopsy unilaterality was the strongest predictor of pathological unilaterality, with a sensitivity of 88.4%, specificity of 34%, and positive and negative predictive values of 28% and 91%, respectively. It seems logical that increasing the biopsy intensity would improve the performance of the biopsy in cancer detection/lateralization. Tsivian et al. showed that extended versus sextant TRUS biopsies had better specificity (53.9% vs. 37.1%) and negative predictive value (95.8% vs. 91.8%) (Tsivian et al., 2010). On the other hand, Abdollah et al. (2011) showed that at 24-core TRUS biopsy, specificity in predicting unilateral disease remained at 49.4% and positive and negative predictive values at 22.6% and 98.9%, respectively, suggesting that there was a limitation strategy of increasing the number of biopsy cores transrectally. This may be due to the inherently random nature of the TRUS systematic biopsy, as well as poor access to or lack of standard methodology to intentionally sample the anterior zone of the prostate gland through a transrectal approach.

In the 1990s, many operators turned to transperineal mapping biopsies (TMBs) to overcome these limitations. The transperineal approach allows equal access to the anterior and the posterior prostate gland. The use of a 5-mm grid or a systematic biopsy template ensures a consistent biopsy intensity as well as distribution of biopsy cores. Barzell et al. (2007) showed that staging (TRUS) biopsies had a false-negative rate as high as 47% and sensitivity and negative predictive value as low as 54% and 49%, respectively, compared to TMB in determining suitability for hemiablative cryotherapy. Onik and Barzell reported that among 110 males with unilateral cancer defined on TRUS biopsy, more than half were disqualified after undergoing TMB (Onik and Barzell, 2008).

Several groups have attempted to create predictive models combining biopsy and other clinical parameters to accurately predict cancer laterality. Fu et al. (2012), in a cohort of 1495 males, found that only PSA ≤4 ng/mL and unilateral cancer on biopsy were significant factors in predicting unilateral disease at radical prostatectomy. Gallina et al. (2012) were unable to develop a model in a cohort of 321 males, finding only that the number of positive cores was significant, with an area under the curve (AUC) of 0.52. Truesdale et al. (2010) found that an increased burden on TRUS biopsy (2 or fewer vs. more than 2 positive cores) led to a greater likelihood of experiencing pathological persistence of cancer or biochemical recurrence after focal treatment. Most of the data presented in this section were obtained with the objective of predicting the location of all, including low-risk, cancer. There is a current interest in preferential detection of Gleason pattern 4 and 5 cancers, thought to be the culprits in prostate cancer progression, metastasis, and mortality, rather than Gleason pattern 3 cancers that are suitable for AS and the excessive detection of which may lead to overtreatment.

Advanced Imaging Techniques in Prostate Cancer

Multiparametric MRI (mpMRI)

Improvements in MRI technology, such as high-field-strength magnets, development of MRI contrast agents, and improved MRI protocols, have enhanced the resolution of anatomic imaging and led to the addition of functional imaging, culminating in mpMRI, which has a high diagnostic accuracy. These developments have positioned mpMRI as a key enabler for prostate FT and have been responsible for renewed interest in this area because of mpMRI's ability to do the following:

1. Preferentially detect high-grade lesions.
   Dynamic contrast-enhanced (DCE) imaging using Gadolinium-based MRI contrast is highly sensitive to vascularized lesions and preferentially detects larger and higher-grade cancer foci (Arumainayagam et al., 2013; Rosenkrantz et al., 2012). Diffusion-weighted imaging (DWI) quantifies the mobility of water molecules to highlight regions with high cellular density that are often correlated with the loss of glandular architecture in high-grade cancer (Gupta et al., 2013).
2. Improve the detection of anterior-zone prostate cancers.
   Anterior-zone cancers, accounting for 15-33% of prostate cancers in cystoprostatectomy (Nevoux et al., 2012) and radical prostatectomy series (Al-Ahmadie et al., 2008; Bott et al., 2002), are generally diagnosed at a later stage because of the location being relatively inaccessible to TRUS biopsy. Anterior-zone cancers are readily detected (Komai et al., 2013) and characterized on mpMRI (Ouzzane et al., 2011). If done so at an early stage, these are ideal lesions for FT because they are amenable to complete ablation while sparing both neurovascular bundles (Tay et al., 2016a).
3. Improve the detection of EPE.
   Contemporary mpMRI series report an accuracy of 62-74%, sensitivity of 35-63%, specificity of 73-91%, positive predictive value of 36-85%, and negative predictive value of 57-92% in the detection of EPE (Cerantola et al., 2013; Feng et al., 2015; Kayat Bittencourt et al., 2015; Raskolnikov et al., 2015; Somford et al., 2013). mpMRI staging has also been shown to outperform the Partin tables in predicting pathological EPE (AUC 0.88 vs. 0.70) (Gupta et al., 2016). mpMRI also contributes to cancer staging within the limited view of the pelvis.
4. Accurately identify the IL.
   Le et al. (2015b), evaluating 122 patients with pre-prostatectomy mpMRI, reported that although the overall mpMRI sensitivity for tumor detection was 47%, this increased to 72% for larger (>1 cm) tumors, 72% for higher-grade (Gleason ≥7) tumors, and 80% for index tumors. Baco et al. (2015), in an analysis of 135 males, demonstrated that the concordance between the index tumor location on mpMRI and radical prostatectomy specimens was 95%. Delongchamps et al. (2015), evaluating 125 males undergoing presurgical MRI-fusion biopsy, reported that mpMRI failed to detect 95 cancer foci, of which the 14 that were clinically significant harbored only 1 IL, with the remainder harboring <20% pattern 4.

Despite the vast improvements brought by the advent of mpMRI, it is important to note that the technique remains imperfect. The PROMIS study (Ahmed et al., 2017), comparing mpMRI to a 5-mm grid TMB, reported a sensitivity for clinically significant cancer of 93%, specificity of 41%, positive predictive value of 51%, and negative predictive value of 89%. The strength of this study was that it encompassed 11 centers with varying types of equipment and levels of radiologist expertise. Indeed, mpMRI performance is subject to technical variations such as magnetic coil strength (3 tesla is more accurate than 1.5 tesla) (Chang et al., 2008), the use of an endorectal coil (recommended to augment field strength when using 1.5 tesla) (Hricak et al., 2007), and the scanning protocol. Accurate IL identification (Gaziev et al., 2016) and interpretation of EPE (Tay et al., 2016b) are heavily dependent on reader experience. The PROMIS data suggest that mpMRI lesions require histologic confirmation, systematic biopsies of mpMRI negative areas may not yet be omitted,

and a treatment approach heavily reliant on mpMRI may result in a small but not insignificant number of outfield disease.

Complete treatment of the IL would require an accurate estimation of its three-dimensional (3D) volume with the addition of a safety margin. Histo-radiologic correlation studies consistently show that mpMRI underestimates histologic tumor volume by 5-30%, after accounting for shrinkage and other artifacts (Baco et al., 2015; Cornud et al., 2014; Le Nobin et al., 2015; Priester et al., 2017; Radtke et al., 2016; Rud et al., 2014; Turkbey et al., 2012). Furthermore, Priester et al. (2017) reported that the mean axial diameter of an IL is underestimated by 11 mm. Aslim et al. (2021) showed that if only pattern 4 and 5 areas of the lesion were taken into account, a 5-mm treatment margin would be able to completely ablate >95%, and a treatment margin of 6 mm, 98.6%, of clinically significant cancer. Nonetheless, this poses significant challenges for lesion-based ablation, and complete ablation may ultimately require hemi- or zonal ablation.

MRI-Targeted/MRI-Fusion Biopsy

Because of the high sensitivity but low specificity of mpMRI, histologic confirmation of suspicious lesions remains critical. The American College of Radiology and European Society of Uro-radiology have jointly developed a grading system known as *PI-RADS* (Prostate Imaging – Reporting and Data System) (Barentsz et al., 2012), which is now in its second iteration (ACR, 2015). This 5-point Likert scale primarily aids radiologists in identifying clinically significant cancer, and data suggest (Jordan et al., 2018) that as the Prostate Imaging Reporting and Data System (PI-RADS) score increases, the higher the Gleason score of the lesion tends to be. Nonetheless, at this point, histologic confirmation of the Gleason score as a marker of lesion aggressiveness remains important in clinical decision making. A 2016 consensus on patient selection for prostate FT (Tay et al., 2017a) defined Gleason 3+4 lesions as the "sweet spot" for FT. With higher-grade lesions, there is a concern about increasingly aggressive cancer portending a higher likelihood of EPE outside an image-guided target zone that may need to be met with more radical types of treatment.

In-bore MRI-guided biopsy is performed using live MRI guidance within the MRI gantry itself. Such procedures carry high accuracy but are time-consuming and costly, require specialized MRI-compatible equipment, and tend to be available only in academic and select centers.

MRI–ultrasound fusion biopsy is a technique where the biopsy is performed in a standard ultrasound-guided fashion but with the MRI targets overlaid onto the ultrasound image. There are now several commercial systems (Kongnyuy et al., 2016) offering various approaches (transrectal or transperineal) to MRI-fusion biopsy. These typically create an MRI 3D prostate (MRI segmentation) and an ultrasound 3D prostate (ultrasound segmentation), which are then overlapped (registration) to translate the mpMRI lesions onto the ultrasound model using either rigid or elastic deformation to match the two. The probe/needle guide is then navigated toward the lesion by the operator using mechanical, visual, or combined assistance, and the biopsy is performed. However, software fusion has a learning curve as well because the overlay of the MRI image onto the ultrasound image is often imperfect, and the user has to compensate accordingly to achieve an accurate biopsy (Tay et al., 2016c). A matched-pair comparison (Kaneko et al., 2023) of patients undergoing both MRI-targeted transperineal and transrectal biopsies at the same time has shown no differences in clinically significant cancer detection rate.

Another approach is to perform visual estimation targeting (VET), or cognitive/mental fusion, where the operator estimates the position of the suspicious lesion based on its appearance on the mpMRI. In expert hands, this has been shown to be as good as software fusion (Wysock et al., 2014). Ultimately, tumors can only be identified accurately for FT by an experienced multidisciplinary team of radiologists and urologists. Such a team, with members who will ascend the learning curve together, is a critical component of an FT program.

Role of the Systematic Biopsy

Despite advances in MRI-fusion biopsy, experts still agree that a concurrent systematic biopsy is necessary to identify MRI-invisible, clinically significant cancer (Tay et al., 2017a). However, the field remains divided on the intensity of biopsy that is required. Lee et al. (2022) demonstrated, in a cohort of patients undergoing systematic saturation biopsy of an average of 22 cores in addition to MRI-fusion biopsy, that reducing the number of systematic cores to an average of 15, 11, 7, or 5 would progressively reduce the detection rate of clinically significant prostate cancer (16%, 13%, 9%, 8%, respectively) compared to performing the full saturation biopsy (21%). It would also significantly change the planning template for FT.

Brisbane et al. (2022) found that 90% of outfield clinically significant cancers were found within 1 cm of the target lesion or within the "penumbra" of the lesion. Their findings suggest that biopsies taken from both within and around the MRI lesion could help better characterize the cancer.

With a limited systematic biopsy, there is a small but significant chance of undetected clinically significant cancer that might go untreated. This may affect biopsy-based oncologic outcomes of FT, but the long-term impact of small-volume untreated clinically significant cancer remains unclear.

Promising Imaging Modalities

Positron Emission Tomography

Positron emission tomography (PET) with choline-based tracers or fluorodeoxyglucose has been investigated for use in prostate cancer staging (Umbehr et al., 2013). Prostate-specific membrane antigen (PSMA) is a type 2 transmembrane protein that is expressed 100- to 1000-fold in prostate cancer cells compared to normal cells. This led to much early interest in the application of radioisotope-labeled molecules linked to PSMA in imaging for prostate cancer. Ultimately, preoperative 111In-capromab pentatide correlated to radical prostatectomy failed to reliably localize prostate cancer for quadrant focal ablation (Mouraviev et al., 2009).

More recently, the development of small molecules targeting the external epitope of PSMA has led to radiolabeled ligands with ^{18}F or ^{68}Ga that are associated with high levels of accumulation within tumor tissue and rapid clearance from non–target tissue, leading to high contrast for improved prostate cancer detection compared to early PSMA-linked radiotracers (Eder et al., 2012). One potential advantage of PSMA PET over mpMRI is that interpretation is not influenced by biopsy-related artifacts such as hemorrhage or inflammation (Eiber et al., 2015). Rowe et al. reported on 13 patients and noted that although mpMRI had a greater sensitivity for prostate cancer, ^{18}F-DCFBC PET had a higher specificity overall and for detection of high-grade lesions (Gleason 8 and 9) confirmed on radical prostatectomy (Rowe et al., 2015). Additionally, although mpMRI interpretation is transiently impaired by early biopsy-related artifacts such as hemorrhage or inflammation, PSMA PET does not appear to have such a limitation. Bonnal et al. demonstrated that it was feasible to fuse both choline-PET and mpMRI targets to 3D-TRUS to take advantage of the excellent anatomical resolution of mpMRI, as well as functional PET information, to define a target for biopsy (Bonnal et al., 2017).

Multiparametric Ultrasound

Because of the resource intensity of mpMRI along with the limitations of fusion (Tay et al., 2016c), there is now significant interest in extending the capabilities of ultrasound. Ultrasound is already a widely available real-time technique that can be performed at the point of care, and multiparametric ultrasound could remove potential pitfalls associated with the MRI–ultrasound fusion process and target registration. Contrast-enhanced (CE) TRUS produces a detailed image of microvascular distribution within the prostate using highly echogenic microbubble contrast agents that are minute enough to flow within capillaries. TRUS elastography evaluates the

distribution of tissue stiffness within the prostate by measuring the properties of a reflected mechanical wave. Further refinements in this area include shear wave elastography, where the mechanical wave is no longer operator dependent but a reproducible signal that is produced by the ultrasound probe itself. Using shear wave elastography, Boehm et al. (2015) reported a sensitivity of 80.9% and specificity of 69.1% for cancer detection using a cutoff of 50 kPa compared to whole-mount examination of radical prostatectomy specimens. Palmeri et al. (2016) explored the use of acoustic radiation force impulse (ARFI) imaging, a technique where the generated waves occur in a synchronized fashion to eliminate interference from subsequent unwanted internally reflected waves, finding a good correlation with mpMRI in a preliminary study of 29 patients.

A multiparametric approach to ultrasound would encompass the strengths of the individual modalities, and this combination of providing functional information may improve the sensitivities of individual modalities by 13-59% (Postema et al., 2015). There is a need for further prospective work in this area. Computerized TRUS (C-TRUS), first explored for quantifying gray-scale TRUS images in the 1990s, is a technique where the ultrasound image is transformed into digital signals for statistical analysis. With the aforementioned advances in ultrasound imaging, a wealth of data is now available for correlation to pathology that could pave the way for multiparametric ultrasound driven by artificial intelligence in the near future.

Contemporary Patient Selection Criteria

There has been a constant evolution in the criteria for patient selection as a result of the rapid advances in knowledge and biologic understanding of prostate cancer in the last two decades since FT was first explored. Indeed, many early FT cohorts included patients with low-risk prostate cancer who would today be considered candidates for AS. Furthermore, there have been few comparative trials between various modalities in prostate cancer because of the highly patient-centric nature of treatment selection. As a result, there is a paucity of data to guide the development of patient selection criteria for prostate FT, and most FT practice has been shaped by a consensus-based approach (Table 163.2).

In a 2009 consensus meeting (de la Rosette et al., 2010), the stated aim of FT was to eradicate all known prostate cancer while preserving uninvolved tissue to spare genitourinary function. Later, a 2010 consensus (Ahmed et al., 2012a) stated that the aim of FT was to treat cancer and leave intact the benign prostate and surrounding normal structures. By 2013 (Donaldson et al., 2015), the consensus was that the aim was to treat the dominant or IL. Here, residual untreated disease was first quantified, with the panel agreeing that it was acceptable not to treat lesions of Gleason grade 3+3 up to a length of 5 mm, though the issue of the type or intensity of biopsy necessary was not addressed. In yet another consensus panel in 2016 (Tay et al., 2017a), a Gleason 3+4 lesion, when capable of being completely ablated, was determined to be the "sweet spot" for FT. This panel also suggested Gleason 4+3 as a maximum grade for FT, with most members reluctant to extend the role of FT to high-grade cancers. In 2019, it was proposed that FT could be utilized in treatment and retreatment to delay or altogether avoid radical therapy (Gross et al., 2019).

Recommendations for the best approach to patient assessment have evolved as well. In the 2010 consensus, the use of mpMRI was recommended only in state-of-the-art centers, whereas in the 2016 consensus, it was recommended that mpMRI should be used to define cancers for FT whenever possible. In 2010 and 2012, TMB was considered state of the art, whereas in the 2016 consensus, MRI-TRUS fusion biopsy was the recommended modality for evaluating a suspicious lesion. Notably, the most recent consensus still recommends a systematic biopsy to evaluate the mpMRI "normal" prostate, recognizing that though mpMRI represents an advancement in prostate imaging, it still has a small but significant false-negative rate.

Sexual function was addressed in the 2016 consensus on patient selection, as well as another 2013 consensus on FT trial design, where the expert panels found that erectile dysfunction was not a contraindication to prostate FT. In the 2016 consensus, although the expert panel agreed that the potential for preservation of erectile function is an important reason to choose FT over other modalities, most of the experts would not deny FT to a patient who was already impotent. This may be due to increasing awareness of the impact of urinary, rectal, or other toxicities that may be associated with the indiscriminate use of whole-gland therapy.

Ultimately, treatment selection for prostate cancer remains a complex interplay of disease characteristics and patient factors. Advances and improvements in the tools available to select patients have improved confidence in the ability to recommend FT to a suitable subset of patients. Further research is needed to determine the most effective strategy to identify patients for FT, keeping in mind the cost and morbidity of treating recurrent cancer that might arise from poor selection.

ABLATION PATTERNS AND CURRENT TECHNOLOGIES

Types of Ablation

Many different ablation patterns have been termed FT, leading to significant heterogeneity within the FT literature. Some of these techniques, such as "hockey-stick"-type ablations, would be more appropriately classified under the umbrella term of *partial prostate ablation* (Fig. 163.3). In an international consensus meeting on definitions in FT, Postema et al. (2016) reported that FT should include treatment of a lesion, quadrant ablation, and hemi-ablations. Some authors would further argue that treatment of a lesion can be further classified into image-targeted lesion ablation, where an image-detected, biopsy-confirmed lesion is treated, versus non-image-targeted, whereby the lesion is treated based only on biopsy information.

Ablation Approach

The prostate can be accessed for ablation via a transrectal, transperineal, or less commonly, transurethral approach. The approach chosen will depend largely on:

1. The location of the tumor
2. The desired ablative technology that is available
3. Physician experience
4. Any other anatomical consideration or preferences unique to the patient

Transrectal approaches allow for ease of access to the posterior zone, whereas transperineal approaches offer better access to the anterior zone of the prostate. Cryoprobes and electrodes for irreversible electroporation are typically constructed of metal and are at least 17 gauge in size, making them less suited for a transrectal approach. High-intensity focused ultrasound is produced within an ultrasound transducer that, because of the necessary housing of sufficient transducer rings to produce the desired intensity/focal distance (Constanciel et al., 2013), is usually introduced transrectally. Laser energy, on the other hand, is delivered through highly flexible fibers that can be delivered through either transrectal or transperineal routes. The transurethral route can deliver high-intensity focused ultrasound (HIFU) energy or steam vapor. As urologists become conversant with multiple techniques, an à la carte approach may be adopted, where the modality/delivery combination is adapted to the patient/lesion (Sivaraman and Barret, 2016).

Focal Cryotherapy

Cryotherapy is a thermal ablative modality achieving cell kill through the extraction of heat-producing lethal cold temperatures. Cryosurgery kills target tissues via physical and physiologic changes, including mechanical forces resulting in membrane disruption created by intra- and extracellular ice formation, recrystallization, vascular stasis/ischemia, apoptosis, and immune-mediated effects (Baust et al., 2014). Although it has been in use for the past two decades, recent developments have made it increasingly suited for FT (Barqawi et al., 2014; Baust et al., 2014). First, the new third-generation cryoprobes utilizing the Joule Thompson effect to generate lethal ice are as fine as 17 gauge, allowing them to be maneuvered to the appropriate

TABLE 163.2 Summary of Contemporary Consensus on Patient Selection

YEAR OF CONSENSUS	CLINICAL VARIABLES	2010	2012	2015	2017	2019
Author/Publication Year		De la Rosette et al. (2010)	Ahmed et al. (2012a)	Donaldson et al. (2015)	Tay et al. (2017a)	Gross et al. (2019)
Goal of Focal Therapy		Eradication of all cancer	Eradication of all cancer	Eradication of clinically significant cancer	Eradication of clinically significant cancer	Completely ablate cancer and delay or altogether avoid radical therapy, perhaps indefinitely, through treatment, retreatment, and surveillance
Cancer Determination	Overall	TMB	TMB and mpMRI	TMB or systematic + mpMRI	mpMRI + systematic biopsy	mpMRI + targeted and systematic biopsies
	mpMRI	Can be recommended at state-of-the-art centers	Recommended	No clear recommendation	Recommended as far as possible	Recommended
	Biopsy of suspicious lesion	—	To be studied	Recommended	Recommended, MRI-TRUS Fusion Biopsy	Recommended
	Biopsy of nonsuspicious area	—	TMB	Recommended	Recommended, at least 12 core TRUS biopsy	Recommended
Disease Factors	Risk group	Low-to-moderate risk	—	Intermediate	Intermediate	Intermediate
	Maximum grade	—	—	—	4+3	—, but no cribriform or ductal
	Maximum size	T2a clinically or T2b radiologically	—	No agreement	Up to 1.5 mL or 3 mL if not crossing midline; up to 20% of prostate volume or 25% if crossing midline	—
	Residual disease permissible	None	None	3+3 up to core length 5 mm	3+3 up to core length 1 mm	3+3
Patient Factors	Age/Life expectancy	At least 10-year life expectancy	NA	Not a primary determinant; best suited to life expectancy >10 years and not <5 years	According to major guidelines No upper or lower limit whereby focal is contraindicated	—
	Sexual function	Effect unknown, counseling recommended	NA	NA	Important, but not the only indication for choosing focal therapy	—

mpMRI, Multiparametric magnetic resonance imaging; *MRI*, magnetic resonance imaging; *TMB*, transperineal mapping biopsy; *TRUS*, transrectal ultrasonography.

position within the prostate. Second, variable ice length adjustment allows the ice ball to be contoured exactly to the lesion that is to be ablated. Third, the ice edge can be clearly seen using transrectal ultrasound (Onik et al., 1988), CT, or MRI, and thus the extent of ablation and the ablation margin can be monitored and adjusted in real time by the treating urologist. Fourth, the use of a urethral warmer has greatly reduced the risk of urethral complications arising from cold injury to the urethral mucosa (Cespedes et al., 1997; Donnelly et al., 2001; Vellet et al., 1997). Typically, thermocouples are placed within the intended ablation zone or its margin, along vital structures such as the external sphincter, neurovascular bundles, or Denonvillier fascia, to monitor temperatures in real time (Wong et al., 1997). Excerpts of a video depicting the steps of focal cryotherapy in the form of anterior prostate cryoablation are shown in Fig. 163.4.

Tay and Polascik (2016) and Kotamarti and Polascik (2023) reviewed contemporary cohort studies of prostate focal cryotherapy

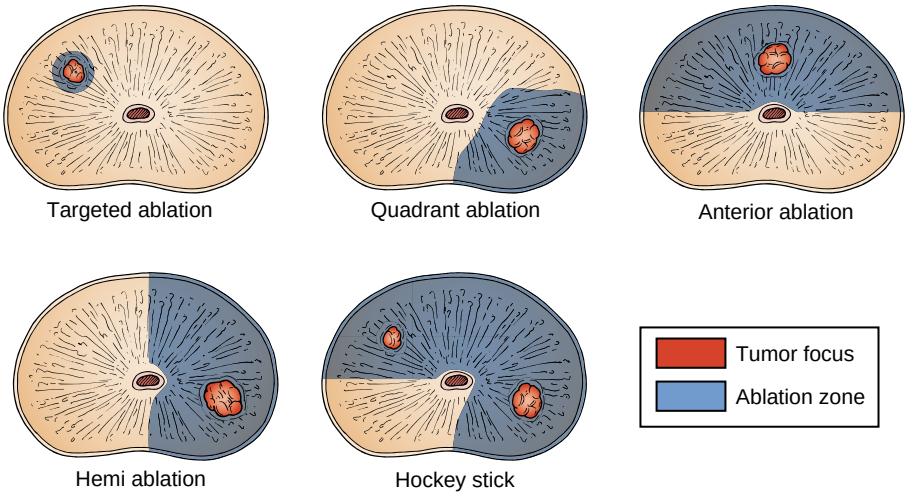


Fig. 163.3. Patterns of ablation.

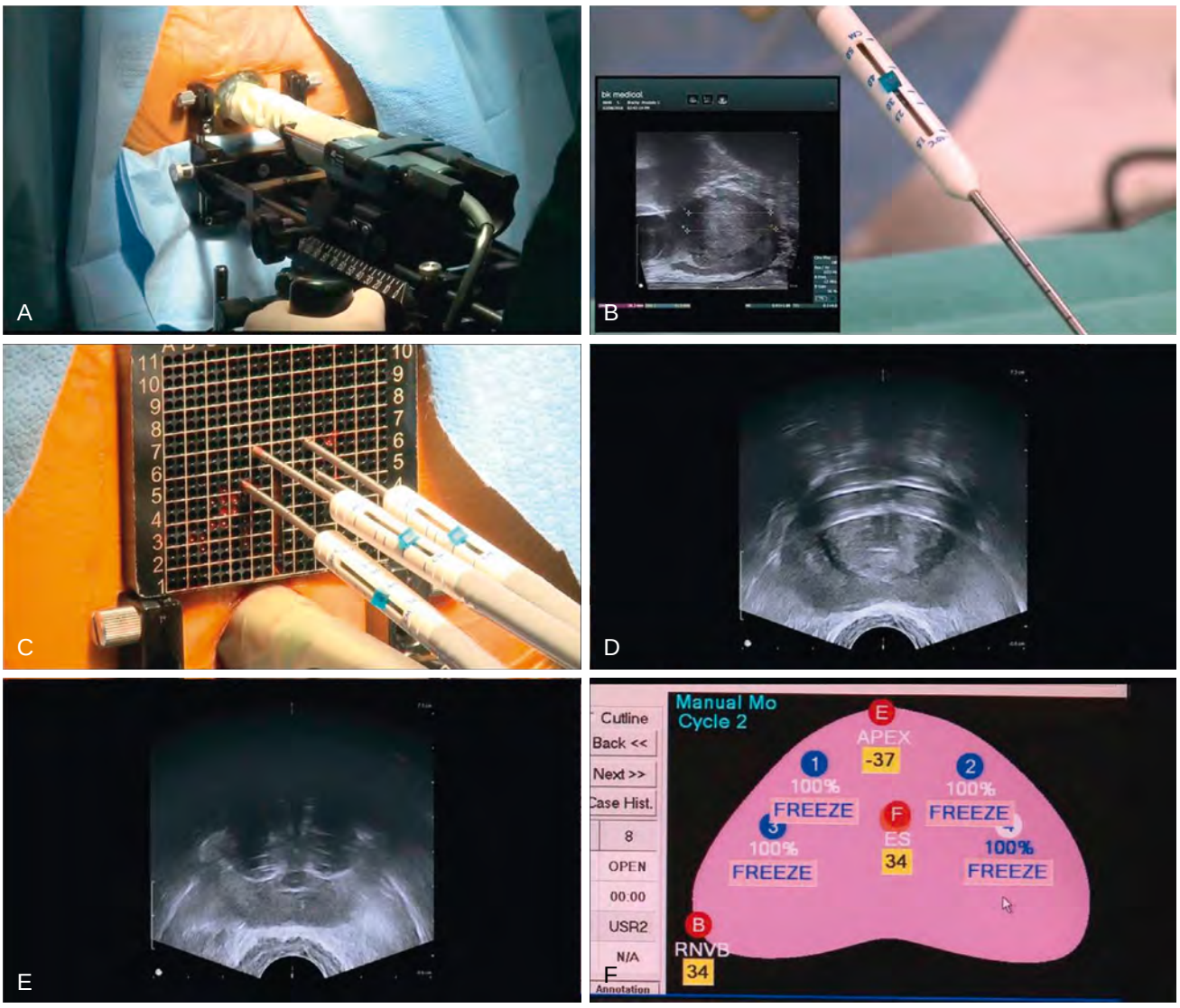


Fig. 163.4. Focal cryoablation procedure. (A) Biplanar ultrasound probe is placed in the rectum. (B) Variable length cryoprobes are set to prostate measurements. (C, D) An anterior ablation pattern is being used here with four probes in the anterior prostate. (E) The ice in the anterior prostate. The leading ice edge has a hyperechoic rim and can be monitored easily. (F) Console readout showing lethal temperature in the anterior apex and safe temperatures in the external sphincter *(ES)* and the right neurovascular bundle *(RNVB)*. The full video may be viewed at http://www.liebertpub.com/videourology. (From Tay KJ, Polascik TJ: Anterior prostate cancer cryoablation: a technique in prostate focal therapy, *Videourology* 31:2, 2017.)

between 2008 and 2023 (Table 163.3). Most studies used a hemiablation template because that was the original model of FT for prostate cancer (Onik et al., 2007a). Over a median follow-up of 6 to 85 months, the biochemical progression-free survival (BPFS) rate was 71-98%, with two studies quoting 3-year BPFS from 56-62.5% (Kongnyuy et al., 2017, 2018) and two studies describing 5-year rates from 62-70% (Mendez et al., 2015; Tay et al., 2017b). Among nine cohorts, which mandated a rebiopsy between 6 and 12 months, the positive biopsy rate was 7.1-38.9%, a quarter of which were clinically significant cancers. As demonstrated in Table 163.3, biopsy criteria for patient selection before therapy and posttreatment biopsy varied across studies. Also, D'Amico risk groups were mixed, but most of the patients treated with focal cryotherapy were of low- and intermediate-risk groups. The reported posttreatment pad-free continence and potency rates were 95.1-100% and 46.8-100%, respectively. Approximately half of the patients reporting potency following hemiablation utilized phosphodiesterase type 5 inhibitor (PDE5I) medications (Ellis et al., 2007). Complication rates mentioned in the cohorts were all low, and high-grade Clavien ≥3 complications did not surpass 4% in any study. Urinary retention was the most encountered adverse event, likely related to reactive edema, and can generally be avoided by maintaining a urethral catheter for a few weeks. The urinary retention rates were recorded at 0-9% after focal cryoablation, and recto-urethral fistula was seen in up to 0.8% of patients across all cohorts. Other reported complications include hematuria, hematospermia, urethral sloughing, and stricture.

There are no studies comparing focal cryotherapy to established treatments such as radical prostatectomy or radiation. One study (Mendez et al., 2015) compared partial- to whole-gland cryotherapy using a matched-pair analysis in patients with low-risk prostate cancer in the Cryo-Online Data (COLD) registry. No difference was observed in 60-month BPFS, and the rates of erectile recovery were noted to be significantly better with partial ablation (68.8% vs. 46.8%). A second study (Tay et al., 2017b) used propensity score matching to compare partial- to whole-gland cryotherapy in patients with intermediate-risk prostate cancer from the same registry. This study reflects a more contemporary approach to selecting focal cryoablation for patients with intermediate-risk disease. There was no difference observed in early BPFS, and those receiving partial ablation were 2.1 times as likely to recover erectile function by 12 months compared to those undergoing whole-gland cryoablation (46.8% vs. 29.5%, $P = 0.001$).

Focal Salvage Cryotherapy

The management of local recurrence of prostate cancer following definitive radiation therapy is challenging not only for achieving good cancer control but also for reducing the morbidity associated with salvage options. Wenske et al. (2013) evaluated 55 patients treated with focal salvage cryotherapy for radiorecurrent prostate cancer. At a median follow-up of 47 months, they found the 5-year and 10-year BPFS were 47% and 42%, with favorable side-effect profiles. In a retrospective comparative study, Abreu et al. (2013) studied 50 males treated with either focal or whole-gland salvage ablation. They found similar rates of oncologic control in both groups, with better functional outcomes in focal salvage ablation patients. Similarly, Tan et al. (2020) found similar oncologic outcomes of focal ablation in comparison to whole-gland salvage ablation in a large cohort of 385 males. The BPFS rate at 2 years was 74%, with 56% of patients who were initially potent having preserved erectile function. The most worrying complication following salvage therapy, rectourinary fistula (RUF), was reported in 1.4% of the cohort (Table 163.4).

Focal High-Intensity Focused Ultrasound

In HIFU, a transducer focuses multiple ultrasound beams onto a preset point, generating a temperature of at least 55°C (Hildebrandt et al., 2002; Napoli et al., 2013) to produce coagulative necrosis within the desired target. The transducer is typically built into the transrectal ultrasound probe, which is also used for treatment planning and monitoring. Transrectal HIFU is well suited for the treatment of posterior-zone lesions. However, the anterior gland may be more difficult to treat as a result of energy dissipation over the intervening prostate tissue and displacement of anterior-zone targets, with gradual edema of the prostate tissue as treatment progresses (Clyne, 2013; Shoji et al., 2013; Sivaraman and Barret, 2016). Typically, there are restrictions based on which prostates are ideal for treatment matched to a particular HIFU technology, with such limitations as prostate size, height, and calcifications in the treatment zone. Also, to better allow targeting and reduction of prostate height, some would recommend that a TURP be performed before HIFU treatment. Finally, although whole-gland HIFU takes some time to accomplish, it is well suited to FT because the treatment time is shorter.

El Fegoun et al. (2011) initially documented 12 males who had biopsy-guided HIFU hemiablation. Over 127 months of posttreatment monitoring, they observed a 90% 5-year recurrence-free rate and a 38% rate at 10 years. The positive biopsy rate was 41.7%, but this was mainly among low-risk cases and before the MRI era. A comprehensive review of focal HIFU series from 2011 to 2022 was reported (Table 163.5). The majority of focal HIFU series are hemiablative and rely on cognitive or software MRI-fusion. Although the criteria for inclusion differed, most groups involved had cases of localized disease, with D'Amico low- and intermediate-risk cases being the most common. With a median follow-up ranging from 6 to 45 months, the infield positive biopsy ranges from 16.7-54% for all cancers and 0-21.8% for clinically significant cancer, and the outfield positive biopsy rate ranges from 7-42.8% for all cancers and 2-7.5% for clinically significant cancer. The overall reported continence rate ranges from 90-95%. The sexual outcomes were heterogeneously defined and reported, such as mild reduction in International Index of Erectile Function (IIEF) score from baseline and nonvalidated questionnaire-like ability to penetrate in patients who are preoperatively potent, ranging from 50-78% at 6 to 12 months. Of these, the large prospective study of focal HIFU involving 1379 patients reported a failure-free survival of 68% for intermediate-risk prostate cancer and 65% for high-risk prostate cancer (see Table 163.5; Reddy et al., 2022).

MRI-Guided Transurethral Ultrasound Ablation

In-bore MRI-guided focal ultrasound (MRgFUS) allows for monitoring of treatment temperatures within the target area using magnetic resonance (MR) thermometry, as well as to ensure safe temperatures in the urethra and sphincter (Fig. 163.5). Tay et al. (2017c) reported a phase II clinical trial of MRgFUS finding a slight insignificant deterioration of Expanded Prostate Cancer Index Composite (EPIC) urinary symptoms score at 1 month, which returned to baseline at 3 months, with no changes in other EPIC subdomain scores over a 2-year period. Another new development that has arisen as a result of miniaturization of HIFU probes is MRI-guided transurethral ultrasound ablation (TULSA) (Fig. 163.6; Burtnyk et al., 2015). In-bore MRI guidance offers the advantages of thermometry and anatomic resolution, and via a transurethral approach, TULSA could potentially be much more efficiently applied to the anterior prostate. A multicenter phase I clinical trial of whole-gland TULSA was reported by Chin et al. (2016), showing the technology to be safe and technically precise, achieving spatial control of thermal ablation to within 1.3 mm on MRI thermometry, with no posttreatment changes in functional scores and a 3.3% grade 3 complication rate with no grade 4/5 complications. Nair et al. (2021) studied the oncologic outcomes of TULSA on localized low- and intermediate-risk prostate cancer, where the estimated 3-year BPFS and salvage-free survival rate was 74% and 76%, respectively, with a 12-month positive biopsy rate of 34% for clinically significant cancer and 59% for any cancer.

Focal Salvage High-Intensity Focused Ultrasound

Kanthabalan and colleagues (2017) evaluated the medium-term cancer-related outcomes of focal salvage HIFU in 150 males who experienced a cancer recurrence after radiotherapy. Although

xt continued on p. 3527

TABLE 163.3 Summary of Published Focal Cryotherapy Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Onik et al. (2008)	48	TRUS biopsy (restaging TMB after 2001)	Unilateral cancer	Low risk: 23 (48%) Int. risk: 18 (38%) High risk: 7 (14%)	Hemiablation	3 monthly PSA for 2 years, then 6 monthly routine biopsy at 12 months	54	92% 1-year BPFS; 85% absolute BPFS (ASTRO); 8.3% (outfield) positive biopsy	90% of those previously potent retained ability to penetrate	NR
Truesdale et al. (2010)	77	TRUS biopsy	Unilateral cancer	Low risk: 44 (57%) Int. risk: 31 (40.3%) High risk: 2 (2.6%)	Hemiablation	3- and 6-month serum PSA and 6 monthly after 12-core TRUS if BCR (Phoenix)/ positive DRE	24	72.7% BPFS (Phoenix), 3.9% (infield), and 10.4% (outfield) positive biopsy	1.9-point decrease in IIEF at 12 months	NR
Bahn et al. (2012)	73	TRUS biopsy	Unilateral, Gleason 7 or less	Low risk 24 (33%) Int. risk 49 (67%)	Hemiablation	3 to 6 monthly PSA 6- to 12-month TRUS, then yearly	44.4	1.4% (infield) and 15.1% (outfield) positive biopsy	74% had ability to penetrate at 1 year and 86% at 2.4 years	Rectal injury: 0%
Ward and Jones (2012)*	11,601	NR	NR	Low risk: 541 (47%) Int. risk: 473 (41%) High risk: 143 (12%)	NR	NR	21.1	2-year BPFS 75.7%	59% had ability to penetrate at 1 year	Retention: 1.2% Recto-urethral fistula: 0.1%
Hale et al. (2013)	26	Staging TMB	Low-to-intermediate risk	Low risk: 23 (88.5%) Int. risk: 3 (11.5%)	Hemiablation	3 monthly PSA for 2 years, then 6 monthly Triggered TRUS bx if PSA/DRE abnormal	19.1	88% BPFS (using nadir + 0.5); 7.7% biopsy positive	73% needed assistance No impotence reported	Retention needing TURP: 4% (1), UTI: 4% (1) Rash: 4% (1)
Barqawi et al. (2014)	62	TMB	40–85 yr cT1–T2b Gleason 3+4 or less Less than 50% positive core after TMB, <20% total prostate, index lesion <5 cc, 4 or less zones involved	All low risk	Targeted sectoral ablation	PSA at 3, 6, 12, 18, 24 months 12-month mandatory 12-core TRUS biopsy	28	71% BPFS (any postop increase); 19.4% positive biopsy	No change in IIEF at 24 months	NR

Durand et al. (2014)	48	12-core staging TRUS biopsy + mpMRI	PSA <10 Positive cores <33% % core <50 Unilateral Gleason 6 or less	All low risk	Hemiablation	3 monthly for 1st year, then 6 monthly PSA 12-core TRUS mandatory at 12 months	13.2	98% BPFS (Phoenix), 12.5% (in-field), 14.6% (outfield) positive biopsy	Mild reduction in IIEF at 3 months, then back to baseline at 6 months	Retention: 15% Recto-urethral fistula: 2% Cavernous corpus necrosis: 2% Urethral stenosis: 2%
Lian et al. (2016)	41	Minimum 12-core biopsy	Unilateral cancer PSA <20 1–2 cores <50% involvement Gleason 6–7 cT2b or less	Low risk: 23 (56%) Int. risk: 18 (44%)	Hemiablation	3 monthly for 1st year, then 6 monthly PSA 12-core TRUS biopsy at 6–12 months, then yearly/ triggered	63	95% BPFS (Phoenix), 4.9% (infield), and 12.2% (outfield) positive biopsy	76.9% of those previously potent retained ability to penetrate	Retention: 3.4%
Mendez et al. (2015)	317	NR	Gleason score ≤6, PSA level at diagnosis <10 ng/mL, and clinical stage ≤T2a	All low risk	Focal	Reflex biopsy	58.3	5 years BPFS: 71.3%	12 months: 63% had the ability to penetrate	12 months: Urinary continence—100%; urinary retention—1.3%
Kongnyuy et al. (2017)	163	NR	Unilateral localized disease	Low: 85 (52%) Intermediate: 67 (41%) High: 11 (7%)	Hemiablation	PSA checks: year 1—every 3 months, then every 6 months	36.6	3 years BPFS: 56%	NR	NR
Tay et al. (2017b)	166	NR	GGG2, PSA >10–20, clinical stage T2b	Intermediate (included 34.9% GGG1): 166 (100%)	Partial	NR	31	BPFS: 2-year—80.7%; 5-year—70% 2.1% positive biopsy	46.8% had ability to penetrate	Retention: 6.6%; no RUF 12 months: Urinary continence rate — 95.1%
Elshafei et al. (2018)	829	NR	Localized disease	Low: 504 (60.9%) Intermediate: 246 (29.7%) High: 76 (9.2%)	Partial	NR	25.2	BPFS: 2-year—83.2%	NR	NR
Kongnyuy et al. (2018)	104	NR	Unilateral, positive cores < GGG4	Low: 41 (39.4%) Intermediate: 53 (51%) High: 8 (7.6%)	Hemiablation	Every 3 months for a year, every 6 months thereafter for PSA and DRE	19	3-year BPFS: 62.5%	NR	NR
Werneburg et al. (2018)	88	NR	PSA <10, unilateral disease, lack of MRI ECE	3+3 —37 (42.1%), 3+4—39 (44.3%), 4+3—12 (13.6%)	Hemiablation	PSA at 3-month intervals	NR	NR	Mild reduction in IIEF score which recovered by 2 years	EPIC: Initial reduction followed by rise at 12 months

Continued

TABLE 163.3 Summary of Published Focal Cryotherapy Series—cont'd

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Inoue et al. (2019)	5	3D mapping biopsy	MRI visible, targeted biopsy-proven GGG3 localized disease	Intermediate: 5 (100%)	Targeted	NR	6	Median PSA decline rate of 71.4% at 6 months MRI: 6–12 months—0 persistent lesions	NR	NR
Bakavicius et al. (2019)	126	Systematic and targeted transperineal biopsies	Localized disease	3+3: 94 (74.6%) 3+4: 31 (24.6%) 4+3: 1 (0.8%)	Targeted	NR	11	NR	NR	Urinary retention: 7.1% UTI: 3.2% Fistula: 0.8% Hematospermia: 0.8%
Shah et al. (2021)	122	TRUS, TMB + mpMRI	GGG: 2 or 3 or high-volume GGG1 (3+3)	3+3: 12 (9.8%) 3+4: 89 (73%) 4+3: 19 (15.6%) 4+4: 2 (1.6%)	Focal ablation	mpMRI at 12 months; biopsy after BCR	27.8	57% (infield) and 43% (outfield) positive biopsy 3 years: FFS—90.5%	84% of those previously potent retained ability to penetrate	Urinary retention: 5% UTI: 9% Urinary continence:100% at 6 months
Sze et al. (2019)	17	Target and systematic biopsy and/or TMB	Anterior gland tumor	3+3: 12 (71%) 3+4: 2 (29%)	Focal ablation	mpMRI and targeted and systematic biopsy at 12 months	15	No infield recurrence and 20% (outfield) positive biopsy	12 months: Mild reduction in IIEF from 19.5 to 19	NR
Basourakos et al. (2020)	55	MRI/TRUS fusion target and 12-core systematic biopsy	GGG > 2	3+4: 32 (58.2) 4+3: 13 (23.6) 4+4: 7 (12.7) 4+5: 3 (5.5)	Focal ablation	mpMRI and targeted and systematic biopsy at 6 months	6	20% (infield—54% CS) and 17% (outfield: 46% CS) positive biopsy	Mild reduction in EPIC score	Dysuria: 2 UTI: 8
Bossier et al. (2020)	26	Systematic and Targeted biopsy	Low–intermediate risk	Low: 4 (15%) Intermediate: 22 (85%)	Hemiablation	Biopsy after BCR	47	BCR: 25%	54% retained ability to penetrate	NR
Chuang et al. (2020)	61	Systematic and targeted MRI/US fusion biopsy	Unilateral clinically significant GG2 or higher	3+4: 40 (66) 4+3: 15 (25) 4+4: 4 (9)	Hemiablation	mpMRI and targeted and systematic biopsy at 6 and 18 months	18	15% (infield) and 4% (outfield) positive biopsy	Mild reduction in EPIC score	Epididymitis: 2 Urinary retention: 2

Tourinho-Barbosa et al. (2020)	119	Systematic and targeted transperineal biopsy	Low- or intermediate-risk PCa	Low: 79 (66) Intermediate: 40 (34)	Focal ablation	mpMRI at 1 and 12 months; systematic core biopsy at 12 months	45	37% (infield) and 19 (outfield) positive biopsy	NR	Urinary retention: 37 UTI: 28 RUF: 1
Wysock et al. (2021)	83	Systematic and targeted biopsy	Unilateral GG < 4	3+3: 9 (11%) 3+4: 51 (61%) 4+3: 23 (28%)	Focal ablation	mpMRI and targeted + ipsilateral systematic biopsy at 6, 24, and 60 months	6	7% (infield) positive biopsy at 6 months	NR	NR
Tan et al. (2021)	71	mpMRI + targeted and systematic biopsy	Low-intermediate PCa	3+3: 30 (42.3) 3+4: 33 (46.5) 4+3: 5 (7) 4+4: 2 (2.8) 4+5: 1 (1.4)	Focal ablation	mpMRI at 3–6 months, 12–24 months, and 5 years	NR	21.4% infield positive biopsy, PSA decrease at 6 months, MRI ability to predict IFF	Ability to penetrate: 58%	Urinary continence: 100%
Marra et al. (2022a)	121	mpMRI + targeted and systematic biopsy	Low–intermediate PCa	Low: 79 (65.3) Intermediate: 40 (33.1) High: 2 (1.7)	Focal ablation	mpMRI 3–12 months and biopsy at 12 months or earlier for cause	85	38.9% positive (34.6% infield and 36% outfield biopsy)	Mild increase in IIE	Urinary continence: 96% Urinary retention: 10 Hematuria: 6 Epididymitis: 7
Baskin et al. (2022)	75	NR	Unilateral GG >2	3+3: 4 (5.3) 3+4: 44 (58.7) 4+3: 21 (28) 4+4: 4 (5.3) GS9: 2 (2.7)	Focal ablation	MRI and control biopsy at 12 months	20	30.6% BCR	Mild decrease in SHIM score	Hematuria: 2 (2.6) LUTS: 1 (1.3) UTI: 1 (1.3) Perineal pain: 1 (1.3) Testicular swelling: 1 (1.3) Urinary retention: 1 (1.3) Transient incontinence: 3 (4)
Aker et al. (2023)	143	mpMRI + targeted biopsy	Unilateral csPCa: GG2 or more; PSA <20 ng/m:	3+3: 5 (3.5) 3+4: 92 (64.3) 4+3: 36 (25.2) 4+4: 10 (7)	Focal ablation	MRI and MRI-guided biopsy at baseline; 6 and 18 months	NR	35% infield biopsy positive	Mild reduction in sexual score	UTI: 17.5% Perineal hematoma: 11% Hematuria: 9.6% AUR: 8.5% Pain: 4.5%

Continued

TABLE 163.3 Summary of Published Focal Cryotherapy Series—cont'd

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Tan et al. (2023)	28	mpMRI + targeted biopsy and systematic saturation biopsy	GG2, PSA <20 ng/mL, lesion volume <3 mL (single) or 1.5 mL (dual)	3+4 (68%) 4+3 (18%) 4+4 (14%)	Focal ablation	mpMRI + targeted biopsy of the treated zone and new targets/systematic saturation biopsy 12–18 months	18	21% csPCa at 12–18 months: infield only 7.1%, outfield only 10.4%, in- and outfield 3.6%	Impairment in EPIC sexual function domain at 1 month compared to baseline but recovery at 3 months	No Clavien grade ≥3 complications, 43% required urinary catheter up to 5 days

*This was a Cryo On-Line Data (COLD) registry series.

ASTRO, American Society of Radiation Oncology; *AUR,* acute urinary retention; *BCR,* biochemical recurrence; *BPFS,* biochemical progression-free survival; *CS,* clinically significant; *csPCa,* clinically significant prostate cancer; *DRE,* digital rectal examination; *ECE,* extracapsular extension; *EPIC,* Expanded Prostate Cancer Index Composite; *FFS,* failure-free survival; *GG,* Gleason grade; *GGG,* Gleason grade group; *GS,* Gleason score; IFF, in-field failure; *IIEF,* International Index for Erectile Function; *Int.,* intermediate; *LUTS,* lower urinary tract symptoms; *MRI,* magnetic resonance imaging; *mpMRI,* multiparametric magnetic resonance imaging; *NR,* not reported; *pCa,* prostate cancer; *PSA,* prostate-specific antigen; *RUF,* recto-urethral fistula; *SHIM,* Sexual Health Inventory for Men; *TMB,* transperineal mapping biopsy; *TRUS,* transrectal ultrasound; *TURP,* transurethral resection of the prostate; *US,* ultrasound; *UTI,* urinary tract infection.

TABLE 163.4 Summary of Published Focal Salvage Cryotherapy Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Ismail et al. (2007)	100	Bone scan + MRI + TMB	Biopsy-proven radiorecurrent PCa, nonmetastatic	≤6–30 (30), 7–33 (33), ≥8–37 (37)	Focal	Clinical assessment and PSA at 6 weeks, q3 months for first year, 6-monthly thereafter, MRI and biopsy for cause	33, mean	BPFS: 2-yr—72%; 5 years—54%	Ability to penetrate[a]: 14%	Continence rate: 87% LUTS: 16 Urinary retention: 2 RUF: 1 Urethral sloughing: 2
de Castro Abreu et al. (2013)	25	TRUS biopsy + bone scan + MRI/CT	Biopsy-proven radiorecurrent PCa, nonmetastatic	≤6–5 (20), 7–14 (56), ≥8–6 (24)	Hemiablation	PSA and TRUS at 3,6, and 12 months; biopsy at 6, 12, 24, and 60 months	31 (4–90)	BPFS: 5-yr—54.4%	Ability to penetrate[a]: 28%	No incontinence
Wenske et al. (2013)	55	TRUS biopsy + bone scan + MRI/CT	Biopsy-proven radiorecurrent PCa, nonmetastatic	NR	Focal	NR	47 (1.6–230.5)	BPFS: 5 yr—47%; 10 yr—42	NR	RUF: 3 BOO: 1
Li et al. (2015)	91	NR	Biopsy-proven radiorecurrent PCa, nonmetastatic	≤6–41 (45), 7–30 (33), ≥8–16 (17.6), Unknown: 12 (4.4)	Partial	NR	15	BPFS: 2 yr—80%; 5 yr—46.5%	Ability to penetrate[a]: 50%	Urinary retention: 6 RUF: 3 Continence rate: 94.5%
Overduin et al. (2017)	47	MRI + TRUS biopsy	Biopsy-proven radiorecurrent PCa, nonmetastatic	≤6–7 (15), 7–17 (36), ≥8–16 (34)	Focal	PSA at 1, 3, 6, 9, and 12 months; mpMRI at 3, 6, and 12 months; biopsy for cause	24	BPFS: 1 yr—51%	NR	NR
Bomers et al. (2020)	62	TRUS biopsy + MRI/CT pelvic imaging	Biopsy-proven radiorecurrent PCa, nonmetastatic	<6–22 (35.5), 7–23 (37.1), ≥8–10 (16.1)	Focal	PSA at 1, 3, 6, and 12 months; mpMRI at 3, 6, and 12 months; biopsy for cause	NR	BPFS: 1 year—62.6%	NR	RUF: 1
Tan et al. (2020)	72	NR	Biopsy-proven radiorecurrent PCa, nonmetastatic	≤6–31 (45.6), 7–26 (38.2), ≥8–5 (7.4) Unknown: 6 (8.8)	Partial	NR	24.4	BPFS: 2-yr—74%	Erectile dysfunction: 52.6%	Urinary retention: 5.6% RUF: 1.4
Tan et al. (2021)	11	mpMRI + TMB	Biopsy-proven radiorecurrent PCa, nonmetastatic	≤6–2 (18), 7– 6 (54), ≥8– 3 (27%)	Partial	NR	12	3 years: FFS—40%; MFS—50%	NR	RUF: 1

[a]Ability to penetrate of patients who were potent before salvage therapy.

BOO, Bladder outlet obstruction; *BPFS,* biochemical progression-free survival; *CT,* computed tomography; *FFS,* failure-free survival; *Int.,* intermediate; *LUTS,* lower urinary tract symptoms; *MFS,* metastasis-free survival; *mpMRI,* multiparametric magnetic resonance imaging; *MRI,* magnetic resonance imaging; *NR,* not reported; *PCa,* prostate cancer; *PSA,* prostate-specific antigen; *RUF,* recto-urethral fistula; *TMB,* transperineal mapping biopsy; *TRUS,* transrectal ultrasound; *UTI,* urinary tract infection.

TABLE 163.5 Summary of Published Focal HIFU Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
El Fegoun et al. (2011)	12	NR	PSA ≤10 ng/mL, ≤3 positive biopsies with only 1 lobe involved, clinical stage ≤T2a, Gleason score ≤7 (3+4)	Low risk: 10 (83%) Int. risk: 2 (17%)	Hemiablation	3- and 6-month PSA, then 6-monthly 6-monthly clinical assessments for 5 years, then annually 6-core mandatory biopsy at 1 year or if PSA rising	127	90% recurrence-free at 5 years and 38% at 10 years; 41.7% positive biopsy	NA[a]	Retention (8.3%)
Maestroni et al. (2012)	74	NR	All D'Amico risk categories	Low risk: 52 (70.3) Intermediate risk: 12 (16.2) High risk: 10 (13.5)	Focal ablation	mpMRI and biopsy at 6 months	29	BCR: 26.6%	50% maintained potency	Urinary incontinence: 13
Ahmed et al. (2015)	56	mpMRI + TRUS biopsy or TMB	PI-RADS 4/5 with concordant biopsy findings Gleason ≤4+3 PSA ≤20 Stage ≤t3a	Low risk: 7 (12.5%) Int. risk: 47 (83.9%) High risk: 2 (3.6%)	Focal ablation	Clinical review and PSA at 1, 3, 6, 9, 12 months Mandatory targeted biopsy of treated area at 6 months Contrast-MRI at 2 weeks and mpMRI at 6 months	12	34.6% (infield, 11.5% CS) and 7.7% (outfield) positive biopsy	Erections sufficient for penetration decreased from 71.4% at baseline to 58.9% at 12 months	Transient dysuria 16.1% Transient hematuria 64.3% Passage of debris 42.9% UTI 17.9% Required endoscopic procedures 7.2%
Van Velthoven et al. (2016)	50	mpMRI + targeted biopsy	Concordant biopsy with mpMRI Stage ≤T2 PSA <15 Prostate volume <40 mL Life expectancy ≥5 yr	Low risk: 24 (48%) Int. risk: 26 (52%)	Hemiablation	Clinical review and PSA at 1, 3 and 6 months, then 6-monthly TRUS biopsy if PSA recurrence (Phoenix)	39.5	72% (Phoenix) and 64% (Stuttgart) BPFS; 6% (infield) and 10% (outfield) positive biopsy	70% of those previously potent retained ability to penetrate	Retention 8% UTI 6% LUTS 18% Stricture 4%
Feijoo et al. (2016)	71	TRUS biopsy (20 core) + mpMRI	unilateral disease Stage T1c–T2a, Positive biopsies <33% Gleason score ≤3+4 PSA <15 Life expectancy >10 yr	Gleason 3+3: 58 (86.6%) Gleason 3+4: 9 (13.4%)	Hemiablation	Clinical review and PSA at 3, 6 and 12 months, then 6 monthly Mandatory 12-core TRUS biopsy at 12 months	12	15.5% (infield) and 9.9% (outfield) positive biopsy	50% of those preoperatively potent retained IIEF >21	Retention 8.5% (2 required TURP) UTI 8.5%

Rischmann et al. (2017)	111	mpMRI + targeted biopsy	Unilateral cancer, Gleason score ≤3+4	Gleason 3+3: 82 (74%) Gleason 3+4: 29 (26%)	Hemiablation	Clinical review and PSA at 1, 3, 6, 12 months and 6-monthly thereafter mpMRI at 6–12 months and targeted biopsy of suspicious areas	30.4 (mean)	12% (5%) infield and 19% (7%) outfield positive biopsy	78% of 51 patients with good baseline sexual function recovered by 12 months	Retention (5), meatal stricture (1), hematuria (5)
Tay et al. (2017)	12	mpMRI + TMB	Up to two lesions <10 mm Stage <T2b Gleason 3+3 only	Gleason 3+3: 12 (100%)	Sectoral ablation	Clinical review and PSA at 1, 3, 6, 12, 18, and 24 months rTPM at 6 and 24 months mpMRI at 6 and 24 months	24	28.5% infield (21.4% CS) and 42.8% outfield (7.1% CS)	No change in EPIC subdomain score from baseline at 24 months	Transient hematuria or mild LUTS 35.7% UTI 14.2%
Ganzer et al. (2018)	51	TRUS biopsy 12-core + mpMRI	Unilateral disease Stage T1c–T2a, Positive biopsies <30% Gleason score ≤3+4, max core length 5 mm PSA < 10	Gleason 3+3: 43 (84%) Gleason 3+4: 8 (16%)	Hemiablation	Clinical review, PSA, and TRUS 3-monthly Mandatory mpMRI 12-core TRUS biopsy at 12 months	17.4 (mean)	24.6% infield (8.2% CS) and 36.7% outfield (2% CS)	70% had IIEF >17 at 12 months PDE5I use increased from 13.7% to 23.5% at 12 months	UTI (9) No grade 3 complications
Garcia-Barreras et al. (2018)	236	mpMRI + TMB	Gleason <3+4; PSA <15 ng/mL; clinical stage <T2b; unilateral disease	3+3: 188 (79.7) 3+4: 48 (20.3)	Hemiablation	mpMRI and TMB at 12 months	12	22 (infield) and 7 (outfield) positive biopsies	Mild reduction in IIEF-5 (18 to 12.83)	CDC: Grade 1—1; Grade 2—38
Ghai et al. (2018)	8	mpMRI + TRUS	Low–intermediate risk PCa, PSA ≤10 ng/mL, ≤cT2a, and ≤Gleason 7 (4+3)	3+3: 6 (80) 3+4: 2 (20)	Focal ablation	mpMRI and systemic TRUS biopsy at 6 months	6	50% (infield) positive biopsy	Mild reduction in IIEF-5 (50.6 to 44)	Urinary retention: 1 Prostatitis: 1
Annoot et al. (2019)	55	mpMRI + targeted biopsy	Clinical stage <cT2; unilateral cancer; <GG 3+4	3+3: 24 (44) 7: 31 (56)	Hemiablation	mpMRI and systematic and targeted biopsy at 12 months	33	22% (infield) and 18% (outfield) positive biopsy	NR	NR
Glybochko et al. (2019)	42	MRI + TRUS/TMB	Clinical stage <cT2; unilateral cancer; <GG 3+4	<7: 42 (100)	Hemiablation	CEMRI at 6 months and biopsy at 12 months	12	10% (outfield) positive biopsy	Mild reduction in IIEF-5 (20 to 16)	NR

Continued

TABLE 163.5 Summary of Published Focal HIFU Series—cont'd

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Abreu et al. (2020)	100	mpMRI + systematic + targeted biopsy	Unilateral cancer; <cT2; <GG 4	1: 29 (29) 2: 55 (55) 3: 11 (11) 4: 5 (5)	Hemiablation	mpMRI and 12-core systematic and target biopsy at 6–12 months	20	17% (infield) and 40% (outfield) positive biopsy	Mild reduction in IIEF-5 (22 to 21)	Urinary retention: 7 UTI: 5
Nahar et al. (2020)	52	mpMRI + systematic + targeted biopsy	Organ unilateral localized PCa; any ISUP	I 1: 17 (32.7) 2: 24 (46.2) 3: 6 (11.5) 4: 3 (5.8) 5–10: 2 (3.8)	Focal, hemi, quadratic or subtotal	mpMRI and fusion biopsy at 6 and 12 months	12	16.7 (infield) and 13.3 (outfield) positive biopsy	No significant change in IIEF-5	NR
Bacchetta et al. (2020)	32	mpMRI + TMB	PSA ≤ 20 ng/mL; stage ≤ T2c; unilateral PCa	3+3: 14 (42) 3+4: 13 (39), 4+3: 5 (15), 4+4: 1 (3)	Focal, hemi, hockey-stick ablation	mpMRI and reflex biopsy		54 (infield) positive biopsies	NR	NR
Shoji et al. (2020)	90	mpMRI + systematic + targeted biopsy	PSA < 20ng/mL; unilateral PCa	3+3:46 (51) 3+4:18 (20) 4+3:14 (16), 4+4: 12 (13)	Focal ablation	mpMRI at 2 weeks and 6 months 6 months: systematic and targeted biopsy	21	8.9 % (outfield) positive biopsy	Mild reduction in IIEF-5 (43 to 37)	NR
Westhoff et al. (2021)	48	mpMRI	Unilateral disease; PSA <10 ng/mL; GG <2	3+3: 26 (54.2) 3+4: 17 (35.4) 4+3: 5 (10.4)	Focal ablation	mpMRI and biopsy at 12 months	38	26% positive biopsy	Mild reduction in EPIC 26	NR
Dellabaella et al. (2021)	189	mpMRI + transrectal MRI-targeted + systematic biopsy	stage cT ≤ 2, GG ≤3	3+3: 92 (48.7) 3+4: 68 (36) 4+3: 29 (15.3)	Partial gland ablation	mpMRI and biopsy at 12 months	29	11.9% failure	Mild reduction in IIEF	Urinary retention: 32 UTI: 30 RUF: 1
Rompré-Brodeur et al. (2021)	77	mpMRI + transrectal MRI-targeted + systematic biopsy	GGG ≥2, PSA ≤15 ng/mL	3+4 = 48 (62.3) 4+3 = 24 (31.2) 4+4 = 5 (6.5)	Focal ablation	mpMRI and biopsy at 12 months	17	31.2% failure	NR	NR
Ghai et al. (2021)	44	mpMRI + MRI-targeted + systematic biopsy	PSA level of 20 ng/mL or less; GG2 or GG3	3+4: 36 (81.8) 4+3: 8 (18.2)	Focal ablation	mpMRI and biopsy at 5 months	24	7% positive biopsy	No changes in sexual outcomes	NR

Yee et al. (2021)	20	mpMRI + MRI-targeted/systematic biopsy	stage ≤T2; PSA≤20 ng/mL; GG <5	3+3: 12 (54.5) 3+4: 5 (22.7) 4+3: 3 (13.6) 4+4: 2 (9.1)	Focal ablation	mpMRI and biopsy at 6 months	NR	10% outfield biopsy positive	Mild reduction in IIEF (15.1 to 13.7)	NR
Reddy et al. (2022)[b]	1379	mpMRI + transrectal/transperineal systematic biopsy	Gleason score 6–9, stage cT3b or less	Low risk: 84 (6.1) Int. risk: 896 (65) High risk: 386 (28) NA: 33 (2.4)	Focal ablation	mpMRI at 6–12 months; TMB for cause	32	FFS: 7 yr: 69 (64–74)	NR	NR

[a]Majority of patients were not interested in sexual function in this study.
[b]This *was* a HEAT registry series.
BCR, Biochemical recurrence; *BPFS,* biochemical progression-free survival; *CDC,* Clavien-Dindo Classification; *CS,* clinically significant; *EPIC,* Expanded Prostate Cancer Index Composite; *GG,* Gleason grade; *HIFU,* high-intensity focused ultrasound; *IIEF,* International Index for Erectile Function; *Int.,* intermediate; *LUTS,* lower urinary tract symptoms; *mpMRI,* multiparametric magnetic resonance imaging; *MRI,* magnetic resonance imaging; *NA,* not applicable; *NR,* not reported; *PCa,* prostate cancer; *PDE5I,* phosphodiesterase type 5 inhibitor; *PI-RADS,* Prostate Imaging Reporting and Data System; *PSA,* prostate-specific antigen; *RUF,* recto-urethral fistula; *TMB,* transperineal mapping biopsy; *TRUS,* transrectal ultrasound; *TURP,* transurethral resection of the prostate; *UTI,* urinary tract infection.

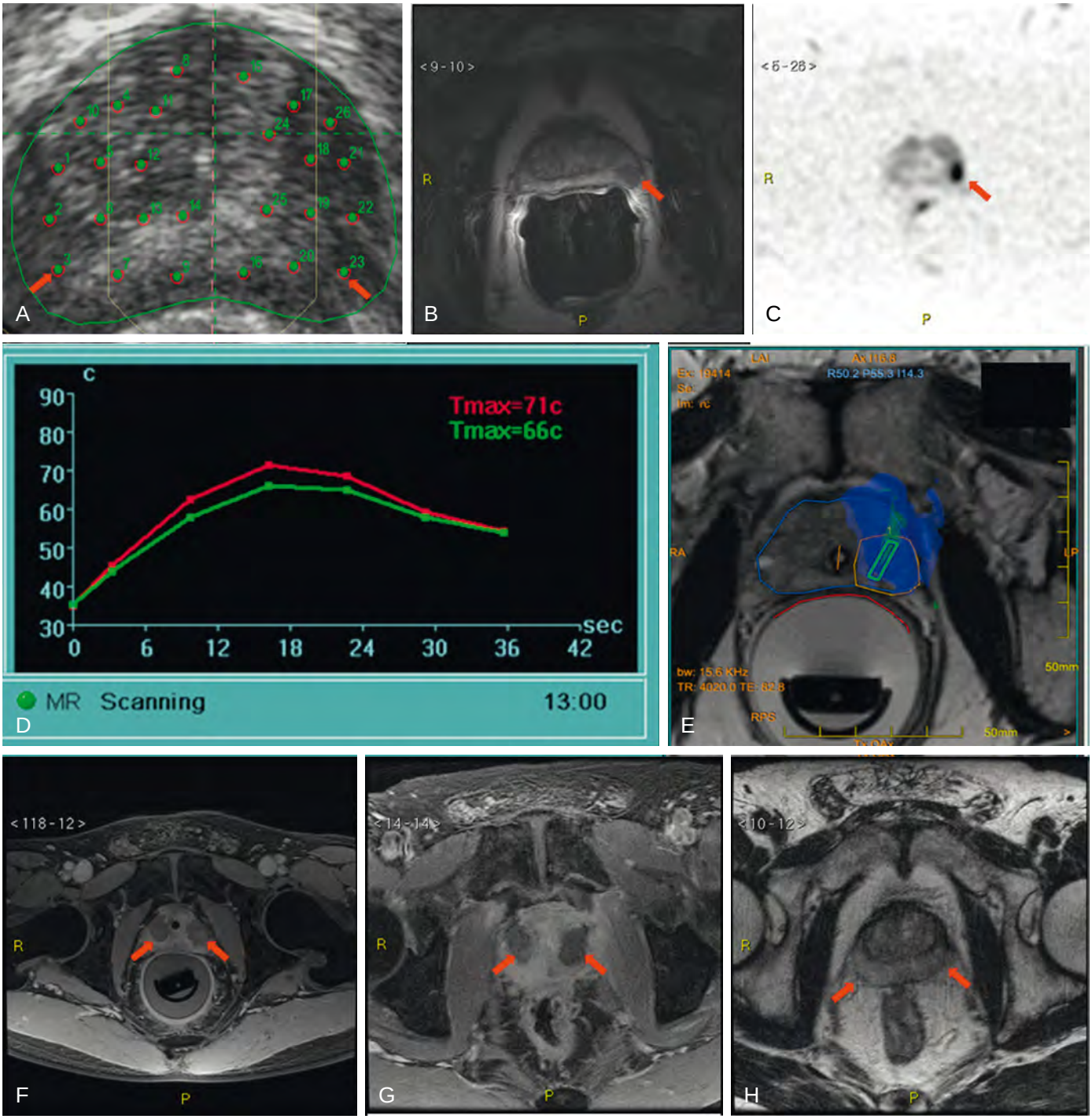


Fig. 163.5. A 54-year-old patient with prostate-specific antigen (PSA) 7.7 and biopsy-proven Gleason 3+3 prostate carcinoma. (A) Saturation biopsy map showing positions of 26 cores taken. Core 3 and 23 were positive for Gleason 3+3 prostate carcinoma *(arrows),* measuring 3 and 4 mm in the mid-cores, respectively. (B, C) Pretreatment axial T2-weighted fast spin-echo MRI (TR/TE, 3300/90) and corresponding axial diffusion-weighted image (DWI) (TR/TE, 105/6000, b = 2000 mm/s^2) showing well-demarcated lesion in left posterior peripheral zone at mid-gland level corresponding to core 23 *(arrow).* It also shows suspicious type 3 curve on dynamic contrast-enhanced (DCE) MRI *(not shown).* The other lesion in the right posterior peripheral zone was not visualized. (D, E) Intraoperative MRI-guided focal ultrasound (MRgFUS) image showing focused ultrasound beam path overlaid on treatment plan for left posterior peripheral zone lesion (E) with real-time thermal map (D). (F) Immediate posttreatment T1-weighted postcontrast MRI (TR/TE, 400/6) showing nonperfused devascularized ablated volumes in bilateral peripheral zones *(arrows).* (G) One-month posttreatment T1-weighted postcontrast MRI (TR/TE, 400/6) showing persistence of nonperfused devascularized ablated volumes in bilateral peripheral zones *(arrows).* (H) Axial T2-weighted fast spin-echo MRI (TR/TE, 3300/90) on 2-year follow-up showing loss of volume with fibrosis in treated area *(arrow).* (From Tay KJ, Cheng CWS, Lau WKO, et al.: Focal therapy for prostate cancer with in-bore MR-guided focused ultrasound: two-year follow-up of a phase I trial—complications and functional outcomes, *Radiology* 285(2):620–628, 2017.)

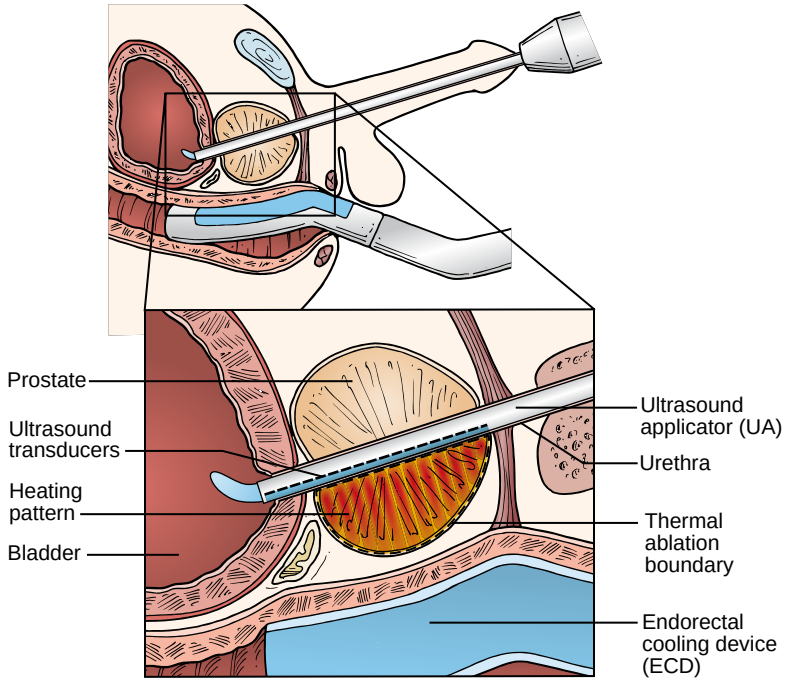


Fig. 163.6. Treatment with MRI-guided transurethral ultrasound ablation. *ECD,* Endorectal cooling device; *UA,* ultrasound applicator. (Chin JL, Billia M, Relle J, et al.: Magnetic resonance imaging-guided transurethral ultrasound ablation of prostate tissue in patients with localized prostate cancer: a prospective phase 1 clinical trial, *Eur Urol* 70(3):447–455, 2016.)

D'Amico risk categories were mixed, most of the patients belonged to intermediate- (23.3%) and high-risk (42%) disease groups before radiotherapy. The reported BPFS and overall survival rate of focal HIFU was 48% at 3 years and 92% at 5 years. Among the cohorts, the reported complications include urinary tract infections (11.3%), bladder neck stricture (8%), and urinary fistula (2%). A subset of around 27% of the patients developed >grade 3, necessitating interventions, whereas 87% remained continent at 2-year follow-up. Focal salvage HIFU therapy seems to offer benefits with relatively low rates of complications and side effects (Table 163.6). The ongoing FORECAST trial will provide further insight into the outcomes of using focal salvage cryotherapy and HIFU.

Focal Laser

Focal laser ablation (FLA) refers to the creation of coagulative necrosis using an interstitially placed laser fiber. The ablation zone is guided by the laser fiber, and tissue destruction occurs only within a fixed distance from the active fiber tip, offering the possibility of true focal ablation where accurate and predictable tissue destruction zones can theoretically be safe within close proximity to vital structures (Oto et al., 2013). The disadvantages are that longer procedure times or extra laser fibers may be necessary for ablation of larger lesions, and additional care may be needed to treat the lesion with an adequate margin. Histopathological analysis of the ablation zone shows neovascularization and increased tumor mitotic activity at the edge, reinforcing the need for a safety margin during treatment (Fig. 163.7; Bomers et al., 2017). The laser fiber can be placed transperineally or transrectally via a delivery trocar.

Reported outcomes regarding the utilization of FLA to date are very small series with short follow-up (Table 163.7). Eggener et al. (2016) treated 27 males in a phase II trial of in-bore MR-guided FLA, reporting an 11% infield and 30% outfield positive biopsy rate at 12 months, with no change in IIEF from baseline. The authors described the use of a 14-gauge titanium introducer with a polytetrafluoroethylene catheter placed in the target lesion, followed by a 17-gauge laser applicator placed through each catheter (Fig. 163.8). Treatment duration was 60 to 120 seconds at 6 to 15 W based on the intended size of the ablative zone and repeated as needed. The target ablation temperature was a minimum of 60°C, with ablation extending more than 5 mm beyond the lesion. Notably, 85% of this cohort comprised low-risk patients.

Natarajan et al. (2017) reported on 10 males with mainly Gleason 7 cancer (80%) undergoing FLA under local anesthesia using a modified MRI–ultrasound fusion biopsy device, where a higher recurrence rate (40% infield and outfield for clinically significant cancer) was observed. They concluded that although FLA is safe and efficient to perform in a clinical setting, close attention should be paid to achieving an adequate treatment margin.

Of these studies, Chao and Lepor (2021) treated 36 low- and intermediate-risk prostate cancer patients with MR-guided FLA and reported a failure-free rate of 83% over a median follow-up of 71 months, with 10/25 (40%) patients developing infield recurrence compared to 25% at 2-year follow-up. They concluded that the high infield recurrence and requirement for salvage therapy following FLA are disconcerting and stressed the importance of close monitoring of these patients and the need for further development in energy delivery.

Although the included studies (see Table 163.7) indicated satisfactory functional outcomes after 6 to 12 months, maintaining continence at a rate of 95-100% and showing minimal impact on IIEF or Sexual Health Inventory for Men (SHIM) scores, recurrence rates varied. Infield recurrence ranged from 11-50%, whereas outfield recurrence ranged from 16.7-60%, observed over a median follow-up period of 6 to 72.5 months. Among the studies, two recorded no severe adverse events (Barqawi et al., 2014; Natarajan et al., 2017). One study noted a single grade III event according to common terminology criteria for adverse events (CTCAEs), which was a urinary tract infection (Al-Hakeem et al., 2019). Additionally, there were 15 grade II events, including two recto-urethral fistulas, both of which resolved within 4 to 6 weeks with the use of a urinary catheter (Walser et al., 2019).

Focal Irreversible Electroporation

Irreversible electroporation (IRE) is a nonthermal ablative technique that uses short pulses of direct-current electricity to produce

TABLE 163.6 Summary of Published Focal HIFU Salvage Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Ahmed et al. (2012b)	39	TMB + mpMRI	Biopsy-proven radiorecurrent prostate cancer, nonmetastatic	Low risk: 6 (15) Int. risk: 13 (33) High risk: 15 (36) Unknown: 5 (12)	Hemi-/Quadrant ablation	PSA q3months for 1 year and q6 thereafter; MRI + TMB for cause	17 (10–29)	BPFS: 2 yr—49%	Mild reduction in IIEF-5 (18–13)	Stricture: 1 Urinary retention: 3 UTI: 3 RUF: 1 Continence: 25 (64)
Kanthabalan et al. (2017)	150	mpMRI + TMB	Biopsy-proven radiorecurrent prostate cancer, nonmetastatic	Low risk: 4 (2.7) Int. risk: 59 (39.3) High risk: 62 (41.3)	Focal: 82 (55) Hemiablation: 51 (34) Index: 11 (17)	PSA q3months for 1 year and q6 thereafter; MRI + TMB for cause	35 (22–52)	BPFS: 3 yr—48%	Ability to penetrate: 7 (58)	Stricture: 12 UTI: 17 RUF: 3 Osteitis: 1 Continence: 42 (87.5)
Baco et al. (2014)	48	mpMRI + TRUS biopsy	Biopsy-proven radiorecurrent prostate cancer, nonmetastatic	Low risk: 10 (21) Int. risk: 12 (25) High risk: 20 (42) Unknown: 6 (12)	Hemi-salvage	PSA q3months for 1 year; mpMRI and biopsy for cause	16.3 (10.5–24.5)	PFS: 2 yr—52%	Mild reduction in IIEF	Osteitis: 2

BPFS, Biochemical progression-free survival; *HIFU,* high-intensity focused ultrasound; *IIEF,* intermediate; International Index for Erectile Function; *Int.*, intermediate; *mpMRI,* multiparametric magnetic resonance imaging; *MRI,* magnetic resonance imaging; *PSA,* prostate-specific antigen; *PFS,* progression-free survival; *RUF,* recto-urethral fistula; *TRUS,* transrectal ultrasound; *TMB,* transperineal mapping biopsy; *UTI,* urinary tract infection.

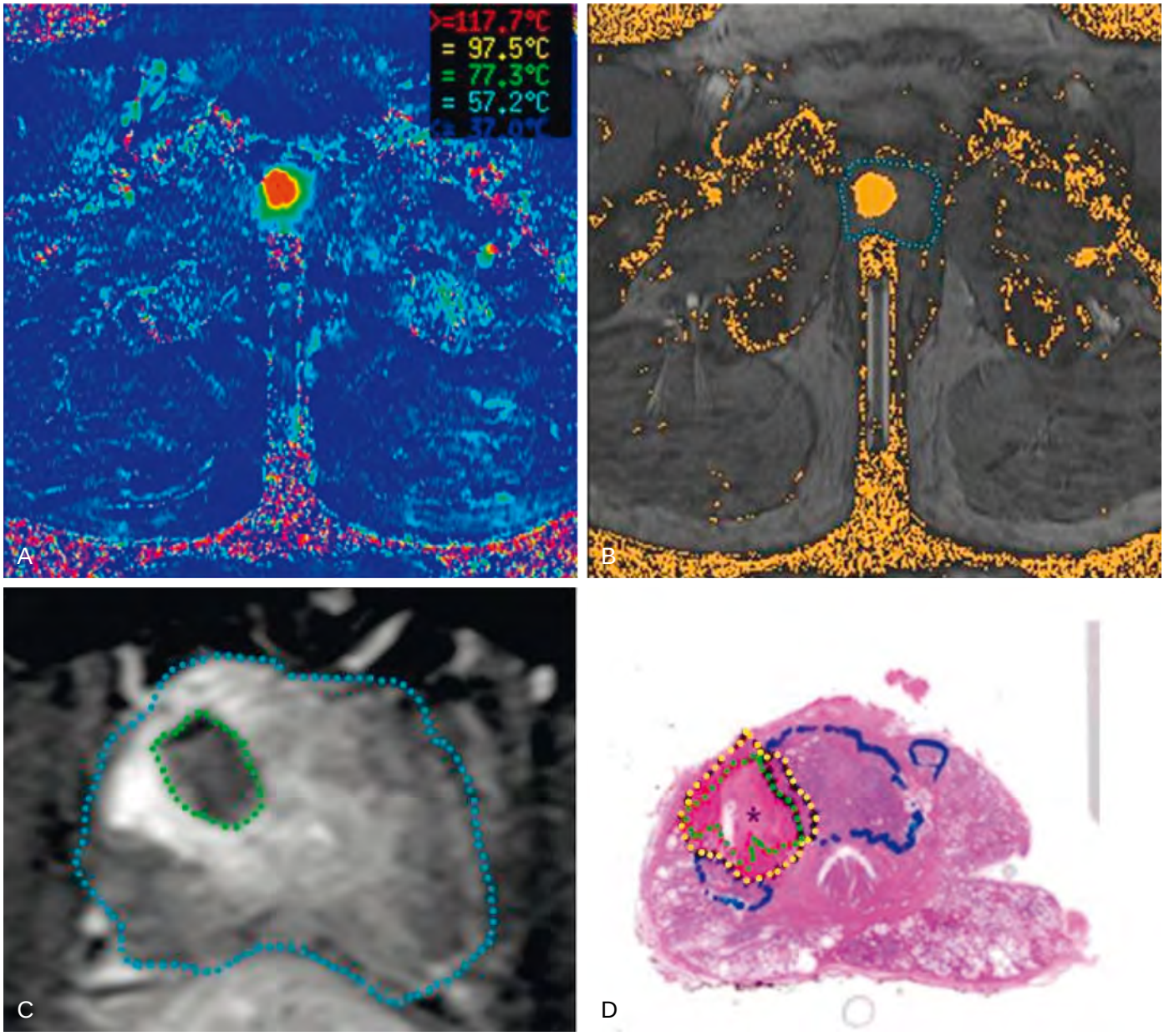


Fig. 163.7. Images of a 67-year-old patient with a Gleason score of 3+4 = 7 in his right transition zone. (A) Real-time MRI temperature map acquired during focal laser ablation. (B) Anatomical MRI of the prostate *(delineated in blue)* with the Arrhenius-based damage-estimation zone as *orange overlay.* (C) Postablation axial T1-weighted contrast-enhanced image. The prostate is delineated in *blue,* and the nonenhancing necrotic tissue is in *green.* (D) Whole-mount hematoxylin and eosin (H&E) stain of the prostate. The necrotic tissue is delineated in *green,* the perinecrotic tissue in *yellow,* and vital tumor in *blue.* (From Bomers JGR, Cornel EB, Futterer JJ, et al.: MRI-guided focal laser ablation for prostate cancer followed by radical prostatectomy: correlation of treatment effects with imaging, *World J Urol* 35(5):703–711, 2017. Open Access via Creative Commons Attribution 4.0 International License: http://creativecommons.org/licenses/by/4.0/.)

irreversible pores in the cell membrane, leading to cell death (Onik et al., 2007b). IRE preferentially damages cells while claiming to preserve connective tissue architecture that could theoretically serve as a scaffolding for new cells to grow. The ablation occurs between two or more probes, typically placed 10 to 20 mm apart (Fig. 163.9). Energy is usually delivered in two components, with the first 20 pulses delivered to characterize the electrical current dynamics between the probe pair. Once adjustments are finalized, the patient is placed in deep anesthesia to avoid muscle depolarization, and the final component of 70 pulses per probe pair is delivered to produce the final tissue ablation. The effective field strength to be achieved is higher than 1600 v/cm between any given probe pair.

There are three early-phase studies (Murray et al., 2016; Valerio et al., 2017; van den Bos et al., 2017) with an aggregate of 104 individuals who have undergone focal IRE for prostate cancer (Table 163.8). One study reported the use of IRE with hemi- or anterior zonal ablations (Murray et al., 2016), whereas the other two treated patients with focal ablation using a template grid (Valerio et al., 2017; van den Bos et al., 2017). Overall, the procedure appears to be safe, with no grade 4 or 5 complications reported. Urinary function appeared to be maintained well, with almost all patients remaining continent, though some reduction in erectile function scores was noted, with slightly fewer patients having erections insufficient for sexual intercourse.

TABLE 163.7 Summary of Published Focal Laser Ablation Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Lindner et al. (2010)	12	12-core TRUS biopsy and mpMRI	≤T2a, PSA <10 ng/mL Gleason sum ≤6 ≤30% of all cores positive ≤50% of 1 core involved by cancer Sectoral correlation with mpMRI	Gleason 3+3: 12 (100%)	MRI-US fusion focal ablation monitored with CEUS	Day 7 mpMRI and CEUS Mandatory 3- to 6-month 12-core TRUS biopsy	6	33% infield positive biopsy and 16.7% outfield positive biopsy	No change in IIEF at 6 months	Perineal discomfort 25%
Eggener et al. (2016)	27	mpMRI	cT1c-T2a PSA <15 ng/mL PSAD < 0.15 Gleason score ≤7 in ≤25% of cores mpMRI max 2 lesions, max diameter 15 mm, max volume 2 mL	Gleason 3+3: 23 (85%) Gleason 3+4: 3 (11%) Gleason 4+3: 1 (4%)	Focal ablation with 5-mm margin	Clinical assessment and PSA at 1, 3, and 12 months 3-month mpMRI with targeted biopsy of infield area 12-month mpMRI and 12-core TRUS biopsy	—	4% infield positive biopsy at 3 months 11% infield and 30% outfield positive biopsy at 12 months	No change in IIEF at 12 months	Hematuria 15% Ecchymosis 11% Retention 8%
Natarajan et al. (2017)	10	mpMRI + MRI-US fusion biopsy	Clinical stage ≤T2b Max Gleason 3+4 Max diameter 2 cm	Gleason 6–2 Gleason 7–8	Focal ablation	Clinical assessment and PSA at 1 week and 1, 3, and 6 months 6-month mpMRI and MRI-US fusion biopsy of target zone, margin, new ROI and 6 template sites	6	50% infield (40% CS) and 60% margin (40% CS)	Median decrease in IPSS 7.5 to 5.5, median decrease in IIEF from 15.5 to 14.5	Hematuria 100%, perineal pain 10%
Chao et al. (2018)	34	mpMRI + Targeted biopsy	Clinical stage <T2a PSA <10 ng/mL Gleason score 8	Gleason 6: 16 (47%) Gleason 7 (3+4): 16 (47%) Gleason 7 (4+3): 2 (6%)	Focal ablation	Clinical assessment and PSA every 6 months mpMRI at 6,12, 24 months MRI-US fusion targeted biopsy at 6 and 24 months	24	25% (18% CS) infield positive biopsy	Mild reduction in SHIM score (22 to 20) at 12 months	100% urinary continence
Walser et al. (2019)	120	mpMRI	Stage T1c–T2b PSA <15 ng/mL PSA density <0.15 ng/mL/cm^3 Gleason score ≤7 No metastases were included	Gleason 6: 37 (30.8) Gleason 7 (3+4): 56 (46.7) Gleason 7 (4+3): 27 (22.5)	Focal ablation	mpMRI and in-bore MRI guidance biopsy for cause	34	18.35 (15% CS) infield positive biopsy	Mild reduction in SHIM score (24 to 22)	RUF: 2 (1.7) Hematuria: 9 (7.4)

Al-Hakeem et al. (2019)	49	mpMRI; targeted and systematic biopsy	cT1c–T2a PSA ≤15 ng/mL clinical stage T1c–T2a PI-RADS score ≥3 on mpMRI Gleason score 7 or Gleason score 6 core length >3 mm.	Gleason 6: 13 (27) Gleason 7 (3+4): 29 (59) Gleason 7 (4+3): 7 (14)	Focal ablation	MRI at 3 + 12 months; MRI/US target + systematic at 6 months	18	20% (infield) and 18% (outfield) positive biopsy	18% maintained penetrance with PDE5I	NR
Chao and Lepor (2021)	30	mpMRI	Clinical stage <T2a disease PSA <10 ng/mL Gleason grade group (GGG) <4	Low risk: 12 (40) Int. risk: 18 (60)	Focal ablation	Clinical assessment and PSA every 6 months mpMRI at 6,12, and 24 months MRI-US fusion targeted biopsy at 6 and 24 months	71.5	FFS: 83%, 40% infield recurrence	NR	NR

CS, Clinically significant; *FFS,* freedom-free survival; *IIEF,* International Index for Erectile Function; *Int.,* intermediate; *IPSS,* International Prostate Symptom Score; *mpMRI,* multiparametric magnetic resonance imaging; *NR,* not reported; *PDE5I,* phosphodiesterase type 5 inhibitor; *PI-RADS,* Prostate Imaging Reporting and Data System; *PSA,* prostate-specific antigen; *PSAD,* prostate-specific antigen density; *ROI,* region of interest; *SHIM,* Sexual Health Inventory for Men; *TRUS,* transrectal ultrasound.

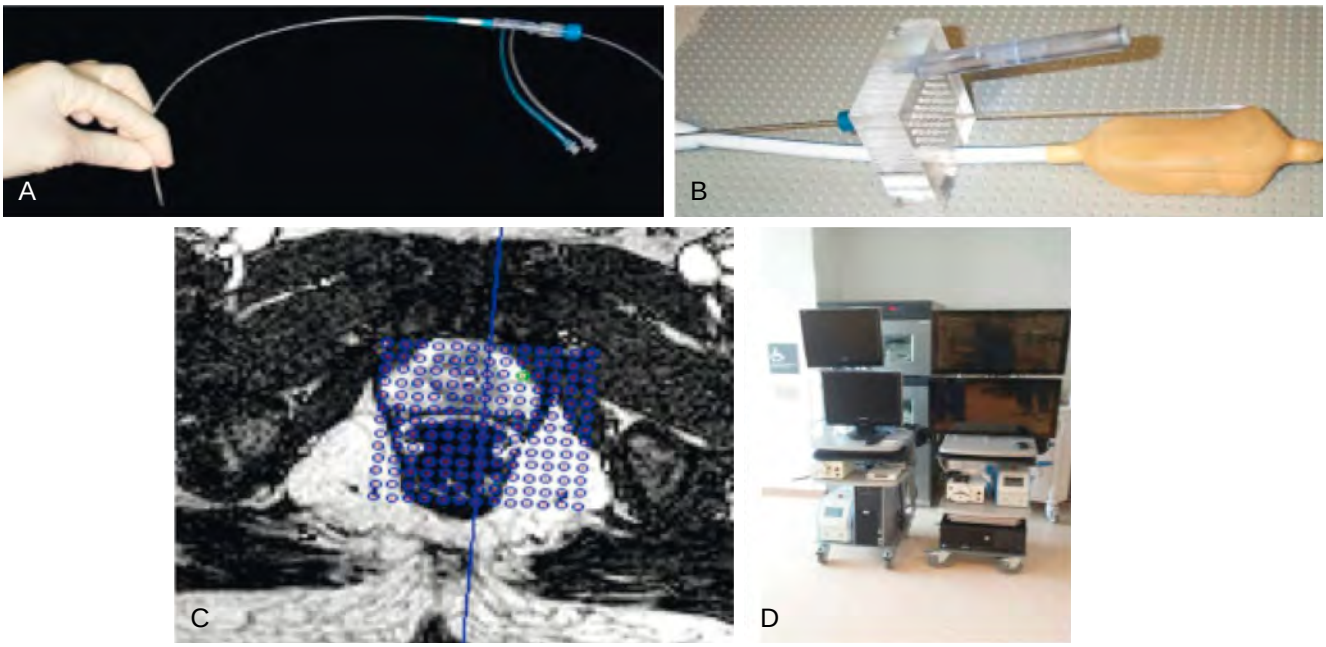


Fig. 163.8. (A) Laser fiber in cooling catheter. (B) Perineal template with endorectal coil and titanium catheter. (C) Overlay of template grid on MRI. (D) Laser workstation. (From Eggener SE, Yousuf A, Watson S, et al.: Phase II evaluation of magnetic resonance imaging guided focal laser ablation of prostate cancer, *J Urol* 196(6):1670–1675, 2016.)

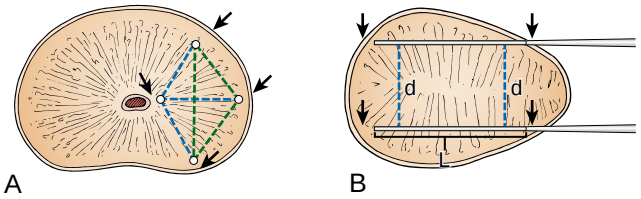


Fig. 163.9. Electrode metrics. (A) Axial representation of probe placement geometry using probe insertion initiated in periurethral position with additional probes configured radially *(blue dashed lines)* from this point. The closest distance from each probe to the capsule or urethra *(arrows)* was no more than 5 mm. The distance between probe pairs *(blue dashed lines)* used for ablation was between 10 and 20 mm. (B) Distances between probes *(d)* and length of needle exposure within tissues *(L)* are obtained to provide data for initial energy settings. Measurements were obtained to confirm parallel orientation *(d)* and a distance of 5 mm or greater tip to capsule *(arrows).* (From Murray KS, Ehdaie B, Musser J, et al.: Pilot study to assess safety and clinical outcomes of irreversible electroporation for partial gland ablation in men with prostate cancer, *J Urol* 196(3):883–890, 2016.)

Further evaluation with larger cohorts is necessary to determine whether the theoretical benefit of connective tissue preservation bears a meaningful clinical effect. Indeed, in a 16-patient ablate and resect study (van den Bos et al., 2016), most of the patients did have fibrinoid necrosis in the unilateral neurovascular bundle and denudation of the prostatic urethra. In this study, the interval between ablation and resection was only 1 month, and whether the remaining connective tissue does allow later regrowth and functional recovery is unknown (Fig. 163.10). Furthermore, an investigation of three recently published larger cohorts (Blazevski et al., 2020; Enikeev et al., 2020; Scheltema et al., 2023) encompassing a total of 451 males, mainly in the low- and intermediate-risk categories, treated with IRE, reveals notable rates of clinically significant infield and outfield recurrence of 1-9.8% and 5-16%, respectively, at 6- to 12-month follow-up. Although a head-to-head trial does not exist, Scheltema et al. (2018) analyzed 50 matched pairs of males who underwent focal IRE or robot-assisted radical prostatectomy (RARP) and found that IRE was significantly superior to RARP in preserving pad-free continence and erections sufficient for intercourse, though the focal IRE patients continued to experience cancer recurrence, with 13 patients having residual clinically significant prostate cancer and requiring salvage therapy (see Table 163.8).

FOCAL SALVAGE IRREVERSIBLE ELECTROPORATION

Geboers et al. (2023) reported the focal salvage outcomes of IRE in 74 individuals with biopsy-proven radiorecurrent prostate cancer. The estimated progression-free survival and metastasis-free survival rates were found to be 60% and 91%, respectively, at a median follow-up of 48 months, with most patients of low- and intermediate-risk prostate cancer. The reported complications were mostly of grade 1 or 2, which included pain, urgency, and mild hematuria. About 9% and 1% of the study population developed grade 3b complications of urethral sludge and rectal fistula requiring intervention, respectively. The reported urinary continence rate was 93%, and 23% had preserved erectile function. Although the study has shown satisfactory oncologic and functional outcomes, this is a single-institution series and requires a large multicenter trial to support the salvage IRE findings (Table 163.9). The ongoing SAFE Pilot trial will provide further insight into the outcomes of using focal salvage IRE (Marra et al., 2022b).

Focal Vascular-Targeted Photodynamic (VTP) Therapy

In vascular-targeted photodynamic (VTP) therapy, low-power near-infrared laser light of a specific wavelength destroys targeted tissues that have been photosensitized with an intravascular agent preferentially taken up by tumor cells (Azzouzi et al., 2013). This occurs via the production of reactive oxygen species in the target zone, causing irreversible damage to cell membranes and small arterioles and leading to tumor necrosis (Vakrat-Haglili et al., 2005). This technology is considered both a drug and a device therapy.

TOOKAD Soluble (WST11, padeliporfin; palladium bacteriopheophorbide monolysotaurine) is a photosensitizer of the bacteriochlorophyll derivatives family. Compared to predecessor

TABLE 163.8 Summary of Published Focal Irreversible Electroporation Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Murray et al. (2016)	25	mpMRI + targeted + systematic 12-core biopsy	Maximum Gleason 3+4	3+3: 18 (72%) 3+4: 6 (24%) 4+3: 1 (4%)	Hemi- or anterior ablations	PSA every 3–6 months mpMRI at 4–6 weeks post-treatment 12-core TRUS biopsy at 6 months	10.9	8% Infield positive biopsy (4% CS), 8% outfield positive biopsy (8% CS)	PQLS urinary symptom score >17, 88% at 12 months from 77% at baseline PQLS erectile function score >22, 65% at 12 months from 59% at baseline	Grade 1 (hematuria, dysuria) 22% 30 d, 0% 90 d Grade 2 (retention, UTI) 26% 30 d and 4% 90 d Grade 3 (UTI) 4% 30 d, 0% 90 d No grade 4–5 events
Valerio et al. (2017)	16	mpMRI + TMB	mpMRI lesions concordant to TMB Untreated secondary cancers insignificant by UCL2 definition allowed	3+3: 8 (42.1%) 3+4: 11 (57.9%)	Focal ablation based on template grid	PSA at 1.5, 3, 6, 9, and 12 months mpMRI and TMB at 6 months	12	38.9% infield positive biopsy (33.3% CS)	100% pad-free/ leak-free at 12 months 69% erections sufficient for penetration compared to 75% baseline	Sloughing (26.3%) UTI (21%) Urethral stricture (5%) No grade 3–5 events
Van den Bos et al. (2017)	63	mpMRI + targeted biopsy + systematic (12-core TRUS or TMB) biopsy	Max Gleason score 7 Single lesion	3+3: 9 3+4: 38 4+3: 16	Focal ablation base on template grid	mpMRI at 1 week and 6 months TMB/saturation TRUS/ targeted biopsy between 6 and 12 months	Min 6	15.6% CS infield positive biopsy 9.8% CS outfield positive biopsy	98% pad-free (same as baseline) Decrease in EPIC sexual function score from 66 to 54 Decrease in ESI from 77% to 53% at 12 months	Grade 1 (hematuria, LUTS, perineal pain): 24% Grade 2 (severe LUTS, UTI, retention): 11% No grade 3–5 events
Scheltema et al. (2018)	60	mpMRI + TMB	Stage < T2b, max Gleason score 8	3+3: 8 (13) 3+4: 40 (67) 4+3: 10 (17) 4+4: 2 (3)	Focal ablation	Transperineal biopsy at 12 months	12	NR	Significant decrease in EPIC sexual score (40 to 27)	NR
Giganti et al. (2019)	30	mpMRI	Organ-confined, biopsy-proven PCa	3+3: 7 (23) 3+4: 20 (67) 4+3: 3 (10)	Focal ablation	mpMRI at 6 and 12 months; cause biopsy	16	10% (outfield) positive biopsy	NR	NR

Continued

TABLE 163.8 Summary of Published Focal Irreversible Electroporation Series—cont'd

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Collectini et al. (2019)	30	mpMRI + TMB/fusion/TRUS	Localized PCa	3+3: 7 (23) 3+4: 4 (24)	Focal ablation	mpMRI at 6 and 12 months; target in-bore biopsy (cause)	20	17.9 (infield) and 6.7 (outfield) positive biopsy	Mild reduction in IIEF-5	NR
Blazevski et al. (2020)	123	mpMRI + TMB/targeted biopsy	Gleason score < 7 and low-to-intermediate-risk PCa; PSA <15 ng/mL	3+3: 12 (10) 3+4: 88 (72) 4+3: 23 (19)	Focal ablation	mpMRI at 6 months; TMB and target biopsy of the ablation zone	36	9.8 (CS) (infield) and 12.7 (CS) outfield positive biopsy at 12 months	Significant decrease in EPIC score (53 to 40)	CDC: Grade 1—22%; grade 2—9%
Enikeev et al. (2020)	12	MRI + fusion biopsy	<T2a; PSA <15, Gleason grade <3	3+3: 11 (92) 3+4: 1 (8)	Focal ablation	mpMRI at 6 and 12 months MRI-fusion and systematic biopsy	12	25% infield (16% CS) and 16% outfield positive biopsy	Mild changes in EPIC score	Hematuria and hematospermia: 4
Wang et al. (2022)	109	mpMRI + transperineal cognitive + systematic biopsy	PSA <20; <T2c; Gleason score of 7 or less	3+3: 47 (43.1) 3+4: 45 (41.3) 4+3: 17 (15.6)	Focal ablation	mpMRI at 1 and 6 months; fusion targeted and systematic biopsy at 6 months	NR	6% (CS 1%) infield and 5% outfield positive biopsy at 6 months	No change in IIEF-5 score (median, 2.0)	Epididymitis (4.6%) Hematuria (3.7%) Urinary retention (2.8%)
Scheltema et al. (2023)	229	mpMRI + MRI-targeted + systematic biopsy	Stage <T2b, max Gleason score 8	3+3: 19 (8.3) 3+4: 159 (69.4) 4+3: 37 (16.2) 4+4: 14 (6.1)	Focal ablation	mpMRI at 6 months and transperineal biopsy at 12 months	60	24% CS: 8% infield and 16% outfield at 12 months FFS: 83% at 5 years	Mild reduction in IIEF-5 from 71% to 58%	NR

CS, Clinically significant; *CDC,* Clavien-Dindo Classification; *EPIC,* Expanded Prostate Cancer Index Composite; *IIEF,* International Index for Erectile Function; *Int.,* intermediate; *LUTS,* lower urinary tract symptoms; *mpMRI,* multiparametric magnetic resonance imaging; *MRI,* magnetic resonance imaging; *NR,* not reported; *PCa,* prostate cancer; *PQLI,* prolonged quality of life symptoms; *PSA,* prostate-specific antigen; *TMB,* transperineal mapping biopsy; *TRUS,* transrectal ultrasound; *UCL,* University College of London; *UTI,* urinary tract infection.

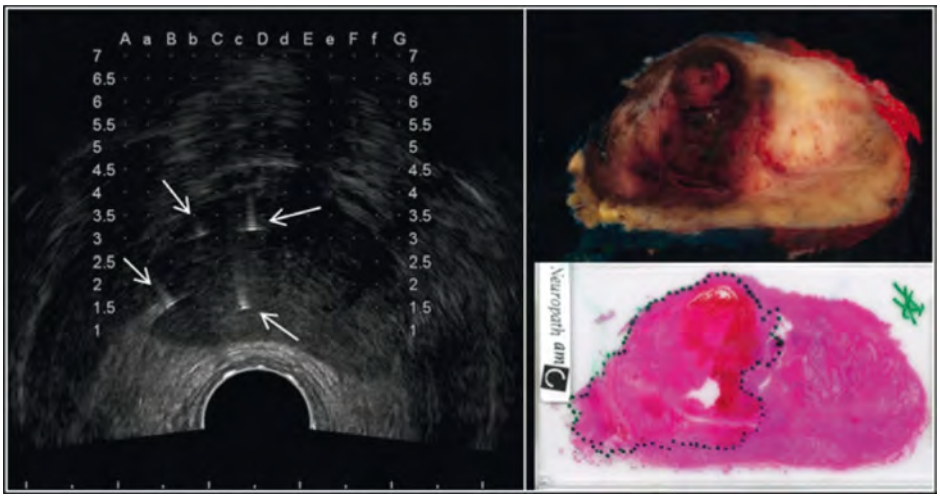


Fig. 163.10. Perioperative transrectal ultrasonography (TRUS) image with inserted irreversible electroporation electrode *(arrows)*, with corresponding whole-mounted prostate section and hematoxylin and eosin (H&E) slide. (From van den Bos W, Jurhill RR, de Bruin DM, et al.: Histopathological outcomes after irreversible electroporation for prostate cancer: results of an ablate and resect study, *J Urol* 196(2):552–559, 2016b.)

agents, it is highly soluble, remains confined to the vasculature even at high doses, and is rapidly cleared by the renal and hepatic systems. Three phase II trials (Table 163.10) using the TOOKAD Soluble photosensitizer for focal VTP of patients with Gleason 3+3 prostate cancer have shown the technology to be safe (Azzouzi et al., 2015). The authors describe a single intravenous administration of TOOKAD Soluble at 4 mg/kg followed by local illumination of the targeted zone using 753-nm laser light at 150 mW/cm and 200 J/cm via a transperineally positioned laser fiber within the prostate. The total anesthesia time was 2 hours, and upon completion, the patients were required to wear protective eyewear and stay in low-level light for the following hour after treatment. A combined analysis of the three trials showed an infield positive biopsy rate of 31.6% overall and 19.4% in those who underwent hemiablation.

Azzouzi et al. (2017) randomized 415 males with low-risk prostate cancer to focal VTP or AS, demonstrating that FT significantly improved the negative biopsy rate at 2 years, decreased disease progression, and reduced the number of those requiring definitive disease intervention. Of the 216 patients assigned to VTP, 196 received the treatment, and 195 completed 24 months of follow-up. In the AS arm, 174 patients completed 24 months of follow-up. Overall, the cancer progression rate (volume, grade, or PSA progression) in the AS arm was 58% compared to 28% in the VTP arm. This is the first randomized trial with level 1 evidence comparing FT to an accepted management strategy demonstrating a meaningful clinical benefit. The study limitations include a Gleason 6 cohort, short follow-up, use of surrogate end points that do not necessarily reflect overall/cancer-specific survival, and that the outcome measure may represent reclassification rather than true progression.

Gill et al. (2018) presented an extended follow-up of 4 years to determine the rate of conversion to radical therapy in ablation and AS cohorts. They found that the ablation cohort was less likely to require radical therapy at 2 years (7% vs. 32%), 3 years (15% vs. 44%), and 4 years (24% vs. 53%) and a decreased disease progression rate. Moreover, the functional outcomes, such as continence rates, were well preserved, with nil to a mild reduction in IIEF score at 24 months in comparison to the AS group. Most patients in the VTP therapy group experienced mild or moderate adverse events, mostly occurring around the procedure or shortly after. These events were swiftly resolved without complications. Pain, attributed to transcutaneous needle placement or prostate swelling, was termed *perineal pain* and was reported by 15% (30 out of 197) in the photodynamic therapy group, compared to less than 1% (1 out of 1) in the AS group.

Focal Radiofrequency Ablation

Radiofrequency ablation (RFA) is a method that utilizes heat energy at approximately 100°C to induce coagulative necrosis in cells. This effect is achieved through molecular friction and agitation, leading to the denaturation of cell proteins and membranes, ultimately resulting in the breakdown of cellular structures. RFA employs a generator that can deliver energy up to 100 W, operating at frequencies between 450 and 1200 kHz. Electrode probes are specially designed for either monopolar or bipolar energy delivery. The integration of thermocouples allows for real-time temperature monitoring during the procedure. A key advantage of RFA lies in its limited conduction through fat, providing a protective shield for adjacent structures. RFA procedures can be guided by either impedance- or temperature-based systems to induce tissue necrosis, with the flexibility to incorporate thermocouples for monitoring localized conditions within the treated area.

Orczyk et al. (2021) studied the early efficacy of bipolar radiofrequency focal ablation in 20 patients with clinically localized low- and intermediate-risk prostate cancer. Of the 21 patients recruited, 20 underwent the procedure, with a follow-up of 12 months. Targeted biopsy at 6 months after treatment showed the presence of clinically significant prostate cancer in 20% of patients. Overall, the functional outcomes were satisfactory at 12 months, with 82% of patients maintaining urinary continence and 92% having the ability to penetrate. Although 40 adverse events were reported during the follow-up period, none were CTCAE grade 3 or greater, with urinary tract obstruction reported in 8 patients, which resolved without any sequelae.

Aydin et al. (2020) reported clinically significant cancer recurrence in the treated area in 30% of patients at 6-month follow-up after focal RFA, with one patient developing a grade 3 complication (gross hematuria) requiring diagnostic cystoscopy without any need for treatment. The urinary continence was preserved in all treated patients, with a significant reduction in IIEF score at 6 months. These results warrant further advancements in technique and close monitoring, as well as attention to ablation margin planning (Table 163.11).

Focal Brachytherapy

Treatment simulation studies (Butler et al., 2016; Haworth et al., 2016) show that an adequate dose delivery to perform FT can be achieved with brachytherapy. The benefit of brachytherapy is the sharp falloff in radiation dose within a few millimeters from the

TABLE 163.9 Summary of Published Focal Salvage IRE Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Geboers et al. (2023)	74	mpMRI + TMB + PET scan	Biopsy-proven radiorecurrent PCa Nonmetastatic	≤6–2 (2.7); 7– 56 (76); ≥8–16 (22)	Focal	PSA-q3 months; MRI at 6 months; TMB at 12 months	48 (31–72)	PFS: 60%	IIEF-5: 7 (23)	Continence rate: 93% Urinary retention: 6% RUF: 1% Perineal pain: 6

IIEF, International Index for Erectile Function; *mpMRI,* multiparametric magnetic resonance imaging; *MRI,* magnetic resonance imaging; *PCa,* prostate cancer; *PSA,* prostate-specific antigen; *PET,* positron emission tomography; *PFS,* progression-free survival; *TMB,* transperineal mapping biopsy.

TABLE 163.10 Summary of Published Focal VTP Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Azzouzi et al. (2015)	117[a]	mpMRI + TMB	PSA <10 ng/mL Gleason pattern 4 allowed at low-volume (<3 cores, ≤3 mm max cancer core length)	Gleason 3+3: 116 (99.1%)	Hemiablation (67) and subtotal ablation (50)	PSA at 1, 3, and 6 months Day 7 and 6-month mpMRI 6-month mandatory 12-core biopsy	—	19.4% positive biopsy for hemiablation (31.6% overall)	Mild reduction in IIEF	Urethral stenosis: 1 Orchitis: 1 Dysuria: 1 Hematuria: 1 Prostatitis: 1 Pelvic pain: 1
Taneja et al. (2016)	30	mpMRI + 12 Core TRUS	PSA ≤10 ng/mL Clinical stage <T2a Unilateral PCa Gleason score <3+3	Gleason 3+3: 30 (100)	Hemiablation	mpMRI and 12 core biopsies at 6 months	12	36.7% infield and 16.7% outfield positive biopsy	No significant changes in IIEF-5	Hematuria: 9 Perineal pain: 8 Hematospermia: 3
Azzouzi et al. (2017)	206	mpMRI + TRUS	PSA <10 ng/mL Clinical stage <T2c Gleason pattern 3 allowed at low-volume (<3 cores, >3 mm to <5 mm max cancer core length)	3+3: 206 (100)	Hemiablation	TRUS 12 core systematic biopsy at 12 and 24 months	24	24% infield positive biopsy	No significant decrease in IIEF-5 score	Hematuria: 55 Urinary incontinence: 17 Dysuria: 51
Lebdai et al. (2017)	82	mpMRI	PSA ≤10 ng/mL Clinical stage <T2a Unilateral PCa Gleason score <3+3	2+3: 1 (1) 3+2: 4 (5) 3+3: 77 (94)	Hemiablation: 61 Subtotal: 21	TRUS 12 core systematic biopsy at 12 months	68	24% infield positive biopsy	Mild reduction in IIEF-5 (23–22.5)	Hematuria: 12 Hematospermia: 3 Perineal pain: 9
Gill et al. (2018)	147	mpMRI + 12 core TRUS	PSA ≤10 ng/mL Clinical stage <T2a Unilateral PCa Gleason Score <3+3	Gleason 3+3: 147 (100)	Hemiablation	TRUS 12 core systematic biopsy at 12 and 24 months	48	25% infield (CS—10%) and 19% (CS—6%) outfield positive biopsy	NR	NR
Rodriguez et al. (2018)	81	TMB	PSA <20 ng/mL Gleason pattern 3 or low-volume secondary Gleason pattern 4	3+3: 69 (85.2) 3+4: 12 (14.8)	Focal ablation	6- and 12-month biopsy	12	NR	Mild reduction in IIEF-5 (19.6–15.8)	Dysuria: 18 Hematuria: 7 Perineal pain: 12
Noweski et al. (2019)	68	mpMRI + 12 core TRUS	PSA ≤10 ng/mL Clinical stage <T2a Unilateral PCa Gleason score <3+3	3+3: 68 (100)	Hemiablation	MRI and 6 month-systematic biopsy	42	25% infield and outfield positive biopsy	15.8% retained the ability to penetrate	Hematuria: 1 Urinary infection: 2 Perineal pain: 9

[a]This was a combined analysis of three phase II trials.

CS, Clinically significant; *IIEF,* International Index for Erectile Function; *mpMRI,* multiparametric magnetic resonance imaging; *NR,* not reported; *PCa,* prostate cancer; *PSA,* prostate-specific antigen; *TMB,* transperineal mapping biopsy; *TRUS,* transrectal ultrasound; *VTP,* vascular-targeted photodynamic.

TABLE 163.11 Summary of Published RFA Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Aydin et al. (2020)	10	mpMRI + TMB	Stage ≤2a PSA <10 ng/mL Gleason score 7	3+3: 7 (70) 3+4: 3 (30)	Focal ablation	mpMRI and transperineal biopsy at 6 months	6	30% infield and 20% outfield positive biopsy	Significant reduction in IIEF-5 score	Hematuria: 4 Hematospermia: 3
Orczyk et al. (2021)	20	mpMRI + TMB	Gleason score ≤7	3+3: 2 (10) 3+4: 17 (85) 4+3: 1 (5)	Focal ablation	mpMRI and target TP biopsy at 6 months	12	20% infield positive biopsy	92% retained ability to penetrate	Urethral stricture: 2 Grade 1: 11 (27.5%) Grade 2: 29 (72.5%)

IIEF, International Index for Erectile Function; *mpMRI,* multiparametric magnetic resonance imaging; *PSA,* prostate-specific antigen; *TMB,* transperineal mapping biopsy; *TP,* transperineal.

radiation source. The radiation dose can be further conformed to the desired treatment area by modulating the rate of radioisotope flow through the interstitial catheters when high-dose-rate (HDR) brachytherapy is performed. With seed or low-dose-rate (LDR) brachytherapy, reported urinary toxicity rates are low and seem dependent on pretreatment symptoms and urethral dose (Gutman et al., 2006). On the other hand, HDR brachytherapy can be associated with a urethral stricture rate as high as 8.2% within the first 2 years of treatment (Hindson et al., 2013). These typically occur at the level of the prostatic apex or bulbous urethra.

Among the published focal brachytherapy studies (Table 163.12), Peters et al. (2019) evaluated the mid-term outcomes (median of 4 years) of 30 patients with low- (13%) or intermediate-risk (87%) prostate cancer treated with HDR brachytherapy. The oncologic outcomes are poor, with biochemical disease-free survival of 70% and 33% having clinically significant outfield recurrence. Mild pretreatment erectile dysfunction deteriorated to moderate/severe erectile dysfunction in 50% of patients with no grade 2 genitourinary or gastrointestinal toxicities. The authors concluded that the poor outcomes might be due to poor patient selection. Large-cohort and long-term studies are required to further determine the efficacy of focal brachytherapy. The focal brachytherapy is unique in its follow-up biopsy protocol. The mandatory biopsies are performed at 24 months after the treatment in published series. This delay in confirmation of treatment success makes early comparison between radiation and nonradiation FT outcomes challenging.

Other Modalities

Focal Intensity-Modulated Radiation Therapy

Technical advances in external-beam radiation have made it possible to deliver a highly focused dose within the prostate, sparing vital structures (Vainshtein et al., 2012). In a randomized phase II trial, Vainshtein et al. (2012) failed to notice any significant benefits of urethral-sparing focal intensity-modulated radiation therapy (US-IMRT) on genitourinary toxicity in comparison with standard intensity-modulated radiation therapy (S-IMRT) in 16 patients with low-risk prostate cancer. In addition, the oncologic outcomes of US-IMRT were found to be inferior to the S-IMRT cohort at a median follow-up of 4.7 years. In contrast, Shimizu et al. (2014) reported better BPFS of 100%, 93.8%, and 89.5% for low-, intermediate-, and high-risk prostate cancer, respectively, at 3 years. About 40% and 0.9% of patients experienced grade 1 and grade 2 genitourinary toxicities.

Dosimetric feasibility has been demonstrated in the use of stereotactic body radiotherapy for FT, though there has not been any clinical trial (Kishan et al., 2015). In a pilot trial involving five patients with low- and intermediate-risk prostate cancer, Fredman et al. (2021) studied the feasibility of focal stereotactic body radiation therapy (SBRT) and reported it to have better dosimetry and minimal toxicity profile, with an early favorable PSA response. Although the study showed satisfactory short-term oncologic and genitourinary toxicity, the results are preliminary, and both long-term and multicenter trials are needed to support these findings because toxicities of radiation therapy are more common late in the course of the disease.

Focal Nanoparticles

Another promising modality is the photothermal excitation of nano-shells. These preferentially accumulate in fenestrated perivascular spaces of tumors, causing the preferential death of cancer cells (Stern et al., 2016). This technology is considered both a drug and a device therapy. An open-label multicenter trial treating 45 patients with low- to intermediate-risk prostate cancer utilizing a single dose of AuroLase (Nanospectra Biosciences, Inc., Houston, TX) therapy and the focal excitation of neoplastic prostate tissue via nanoparticle-directed excitation was reported to have a 73.3% success rate at 12 months by postablation biopsy. These promising results open a new field to explore the therapeutic benefit of nanoparticles as an adjuvant in the treatment success of FT of prostate cancer (Rastinehad et al., 2019).

PRX302 is a genetically modified pore-forming protein (aerolysin) that is activated by enzymatically active PSA within the prostate, subsequently causing cell death. A pilot study of 18 patients showed that a single MR-fusion transperineal administration of PRX302 was well tolerated, with two patients having complete histologic responses and seven having partial histologic responses (Shanmugabavan et al., 2017).

NanoPac, sterilized nanoparticles of paclitaxel, is another modality being studied in the phase IIa setting as an intra-tumoral injectable agent (NCT03077659). Sixteen subjects with Gleason scores $\geq$7 received a transrectal ultrasound-guided injection of NanoPac into the IL and its lobe using mpMRI guidance. A radical prostatectomy was performed 4 weeks after the treatment and showed a decrease in the percentage of adenocarcinoma in the prostatectomy core biopsies of six patients. Among seven subjects who received 15 mg/mL, the mean lesion volume (as measured by mpMRI) decreased, as did the PSA density. There were no drug-related serious adverse events (SAEs), including no prostatitis, or dose-limiting toxicities, allowing for dose escalation to the highest concentration (15 mg/mL). Paclitaxel was detected in all prostate tissue sampled from prostatectomy specimens and in lymph nodes of nine subjects$>$30 days following treatment. NanoPac was found to be well tolerated, with no safety concerns at the 15 mg/mL concentration, and provided preliminary evidence of activity. More devices and pharmaceuticals are expected to be trialed in the field of FT.

Adjuvants to Focal Therapy

Many of the proponents of FT believe that a successful treatment will ultimately not hinge on a device alone but rather on a combinatorial approach (Baust et al., 2015, 2018). SBRT strategies routinely utilize a boost directed to the tumor focus within the prostate, with only 5 to 10 external-beam fractions required. A pilot study of 50 patients undergoing SBRT with a boost to the MR-visible tumor showed no biochemical recurrence at 2-year follow-up and an acute genitourinary/gastrointestinal toxicity rate of 8%/2% and late toxicity rate of 6%/0% (grade 3) (Aluwini et al., 2013). The added efficacy of combining a nonradiation focal ablative therapy with low-dose/fraction external-beam irradiation is yet to be determined. In another investigative approach, DOTA-AR, a bombesin-antagonist peptide targeting gastrin-related peptide receptor (GRPr), when conjugated to a radioisotope, has been found to cause tumor growth delay when used in combination with WST11 (TOOKAD Soluble) in GRPr-positive prostate cancer xenographs (Kim et al., 2017).

In focal cryotherapy, there has been early research exploring means to enhance cell killing. The use of androgen deprivation in combination with cryoablation has not been deemed useful to increase cell death (Grossgold et al., 2014). Alteration of the extracellular environment by infusing a solution with a higher-than-physiologic eutectic temperature can enhance cryoablation by promoting extra- and intracellular eutectic ice formation to provide a cell-kill mechanism in addition to intracellular lethal ice (Han et al., 2004). Several studies have shown the benefits of chemotherapeutic agents (e.g., 5-fluoruracil, cisplatin, doxorubicin, mitomycin and bleomycin, vinorelbine, navelbine and tumor necrosis factor [TNF]-related apoptosis-inducing ligand) as apoptosis inducers to enhance cell destruction at milder temperatures in vitro (Baust et al., 2015). TNF-α, a proinflammatory cytokine, is also being explored as an enhancer of cryoablation (Jiang et al., 2008). Vitamin D_3 has been found to improve the apoptotic effects of cryoablation at sublethal isotherms in murine models (Santucci et al., 2011).

At present, there is not a widely applicable and ubiquitous substance that can be utilized to augment device-mediated cell kill, but further research in this area will likely yield additional refinements to techniques in prostate FT.

POST–FOCAL THERAPY FOLLOW-UP

With FT, traditional markers of therapeutic success, such as serum PSA, may be less relevant. After curative radical prostatectomy, PSA is expected to be undetectable, and after radiation therapy, a significant

TABLE 163.12 Summary of Published Focal Brachytherapy Series

AUTHOR	*N*	PREDIAGNOSTIC WORKUP	DECLARED INCLUSION CRITERIA	FINAL DEMOGRAPHIC	ABLATION PLAN	FOLLOW-UP PROTOCOL	MEDIAN FOLLOW-UP (MO)	ONCOLOGIC OUTCOME	SEXUAL FUNCTION OUTCOMES	COMPLICATIONS
Srougi et al. (2017)	41	mpMRI + TMB	Gleason score ≤3+4 PSA ≤15 ng/mL Clinical stage T2a or lower Unilateral tumor	3+3: 33 (80) 3+4: 8 (19)	Focal ablation	NR	24	NR	Mild reduction in IIEF-5 (19 to 17)	Rise in IPSS (4.9 to 6.4)
Mahdavi et al. (2017)	5	mpMRI + TMB	Gleason score ≤3+4 PSA ≤10 ng/mL Clinical stage T2a or lower	3+3: 3 (60) 3+4: 2 (40)	Focal ablation (LDR)	mpMRI at 12 and 24 months and TMB at 24 months	24	No recurrence	No significant change	No toxicity
Graff et al. (2018)	17	mpMRI + systematic and target biopsy	Gleason score ≤3+3 PSA ≤10 ng/mL Clinical stage T2a or lower	3+3: 13 (76) 3+4: 4 (24)	Focal ablation (LDR)	mpMRI and target and systematic biopsy at 12 months	12	No infield recurrence and 6% (CS) outfield recurrence	No significant change	No toxicity
Peters et al. (2019)	30	mpMRI + fusion biopsy	Gleason ≤7 PSA ≤10 ng/mL Clinical stage ≤T2c	3+3: 16 (53) 3+4: 12 (40) 4+3: 2 (7)	Focal ablation (HDR)	PET/CT after BCR	48	33% outfield positive biopsy	50% new onset impotence	Dysuria: 3 Cystitis: 1
Prada et al. (2020)	50	mpMRI	Gleason ≤7 PSA ≤15 ng/mL Clinical stage ≤T2a	3+3: 31 (62) 3+4: 19 (38)	Focal or hemiablation (HDR)	MRI and biopsy (cause)	32	14% BCR and 10% positive biopsy	34% preserved ability to penetrate	No GU or GI grade 2 toxicity
Langley et al. (2020)	30	mpMRI + TMB	Low- or intermediate-risk PCa	3+3: 5 (17) 3+4: 21 (70) 4+3: 4 (13)	Hemiablation (LDR)	mpMRI and TMB at 24 months	24	19% infield and 8% outfield positive biopsy	73% preserved ability to penetrate	NR
Fischbach et al. (2020)	9	mpMRI + systematic and target biopsy	Low- or intermediate-risk PCa	3+3: 4 (44) 3+4: 4 (44) 4+3: 1 (12)	Focal ablation (HDR)	mpMRI at 6, 12, and 24 months Biopsy at 12 months	6	No recurrence on biopsy	Mild reduction in IIEF-5	No toxicity

BCR, Biochemical recurrence; *CS,* clinically significant; *GI,* gastrointestinal; *GU,* genitourinary; *HDR,* high-dose-rate radiotherapy; *IIEF,* International Index for Erectile Function; *IPSS,* International Prostate Symptom Score; *LDR,* low-dose-rate radiotherapy; *mpMRI,* multiparametric magnetic resonance imaging; *NR,* not reported; *PCa,* prostate cancer; *PET/CT,* positron emission/computerized tomography; *PSA,* prostate-specific antigen; *TMB,* transperineal mapping biopsy.

rise is needed to define biochemical recurrence. After FT, however, the expected fall in PSA has been variable, with some studies reporting a 50-80% reduction. One group has reported that nadir PSA + 1.0 ng/dL at 12 months or nadir PSA + 1.5 ng/dL at 24 to 36 months had high sensitivity and specificity in the detection of clinically significant prostate cancer on for-cause biopsy (Huber et al., 2020). Ultimately, the PSA level would depend on the amount of residual prostate epithelia that may continue to grow and is thus less informative as an absolute marker of success or failure.

Similarly, clinical examination of the prostate may be affected by posttreatment scarring and contraction of the original tumor site, making follow-up biopsy less reliable. These posttreatment findings are seen on mpMRI, TRUS, and CEUS, with increasing ablation-zone volume in the early posttreatment period followed by a gradual overall reduction in prostate volume due to ablation-zone fibrosis and deformation (Scheltema et al., 2017).

Compared to traditional treatments, there are additional dimensions to cancer recurrence after FT that affect the downstream management of these patients. In a 2015 Joint International Consultation on Urological Diseases (ICUD)–Society Internationale de Urologie (SIU) project on image-guided FT (Polascik and Amin, 2015), success was defined as the following:

1. Within the treated or infield area as the:
   a. Eradication of the tumor focus(i) in the short term.
   b. Absence of clinically significant cancer in the intermediate to long term.
2. Within the untreated or outfield area: absence of the development of clinically significant cancer. In the short term, any outfield cancer focus likely represents selection failure, whereas in the intermediate to long term, this may represent de novo cancer.

This consultation recommended an mpMRI with mandatory targeted biopsy of 4 to 6 cores in the treated area at 3 to 6 months, then mpMRI at 1 to 2 and 5 years with targeted biopsy as needed, especially if a new lesion should become manifest. A systematic biopsy was also recommended at 1 to 2 and 5 years to provide further histologic assessment of the untreated zone for potential outfield de novo or progressive disease if low-volume, low-grade disease was present and being monitored.

It should be noted that there is limited experience with mpMRI in the post–focal treatment arena, and it is unknown how posttreatment changes interact with mpMRI performance. In one whole-gland HIFU series, the sensitivity and specificity of DCE-MRI were reported to be 73-87% and 73-82%, respectively (Punwani et al., 2012). As radiologic experience in the post-FT prostate accrues, there will be much for practitioners to learn, especially in those patients experiencing treatment failure.

Prostate imaging after focal ablation (PI-FAB) has been proposed as a novel imaging scoring system that rates mpMRI sequences in sequential order (Fig. 163.11) to guide clinicians on further follow-up (Giganti et al., 2023). PI-FAB proposes the assessment of DCE sequences first because posttreatment scarring and fibrosis often obscure the T2W and DWI sequences that are heavily relied on for treatment-naive PI-RADS scoring. Enhancement of an area >3 mm within the treated zone is considered an indication for biopsy. These criteria are developed based on the experience of the authors, and clinical correlation is presently still recommended until further validation is available. Nonetheless, this represents a step toward standardized image assessment after FT.

Because of the intensity of follow-up, investigators have also sought alternative imaging modalities to mitigate the cost of

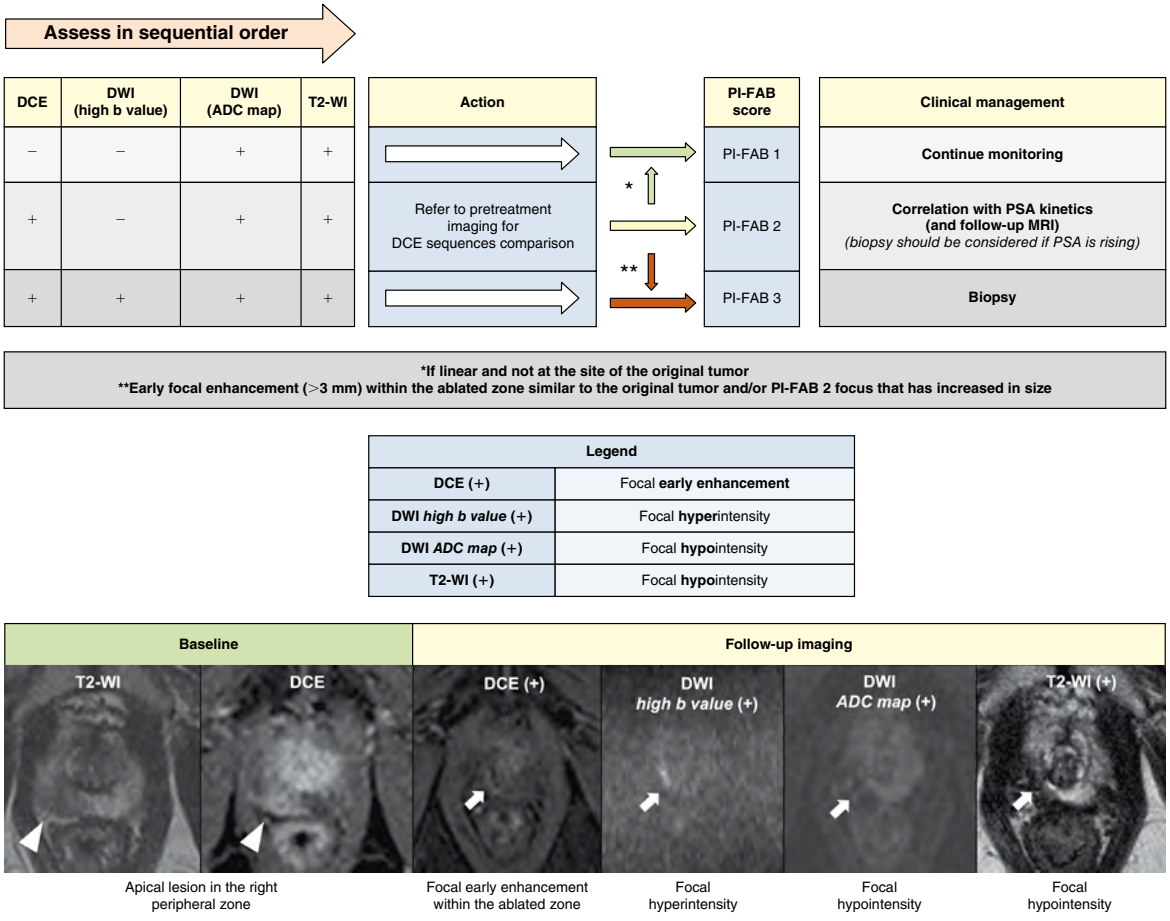


Fig. 163.11. Prostate imaging after focal ablation (PI-FAB) scoring system to determine the likelihood of local recurrence on multiparametric magnetic resonance imaging (mpMRI). *ADC*, apparent diffusion coefficient; *DCE*, dynamic contrast enhanced; *DWI*, diffusion-weighted imaging; *PSA*, prostate-specific antigen; *T2-WI*, T2-weighted imaging.

mpMRI and the morbidity of repeated biopsies. One group analyzed 16 individuals in a pilot study of focal IRE before radical prostatectomy (van den Bos et al., 2016) and found that CEUS was as good as mpMRI in determining early ablation success, whereas TRUS was not able to visualize the ablation zone after treatment. This is an important area for further research.

CONCLUSIONS AND AREAS FOR FUTURE RESEARCH

"Is cure possible when necessary and necessary when possible?"

—Willet Whitmore

Better health care leading to longer life expectancy, an increasing awareness of prostate cancer, and widespread use of PSA screening have led to more prostate cancers being diagnosed. Meanwhile, advances in imaging and biopsy have resulted in these cancers being diagnosed at an earlier stage where, in a subset of patients, FT may render the cancer cured or at least controlled for a significant duration. Because the biology of prostate cancer, for the most part, is one of a slowly growing disease, the management of prostate cancer has the potential to convert this from an acute to a chronic disease. FT, with its combination of eradicating clinically significant cancer while monitoring uninvolved parenchyma on an AS protocol, can avoid overtreatment and preservation of genitourinary function for many years. Many individuals today equally value quality of life as much as cancer control, and image-guided ablation is well positioned to achieve both functional metrics and cancer control. However, the journey remains filled with challenges, and FT remains a work in progress with significant room for further research (Tay et al., 2017d).

First, whether satellite, insignificant cancers can remain safely untreated remains unknown. However, this conundrum is also applicable to patients managed with AS and is not unique to FT. Present efforts have focused on identifying the IL, or the lesion with the highest stage/grade within the prostate, for ablation. However, small or insignificant cancers at the time of treatment may progress to a more aggressive phenotype. Indeed, the only randomized trial of prostate FT compared to AS (Azzouzi et al., 2017) found a progression-free survival benefit in patients with low-risk prostate cancer, suggesting that there is some value to treating lower-grade lesions. Further research is needed to develop clinical, imaging, and genetic biomarkers that can profile all the truly aggressive lesions within the prostate. Techniques focusing on functional and metabolic imaging in addition to anatomic imaging may be an area where such profiling can converge such that genetic and metabolic markers can be detected by imaging for highly accurate and appropriate image-guided ablation. In the meantime, continued close surveillance of any small, insignificant lesions remains necessary.

Second, the long-term cost-effectiveness of FT remains unknown. Up to half of patients on AS for prostate cancer eventually undergo radical intervention (Hamdy et al., 2016; Klotz et al., 2015). By restricting treatment to the clinically significant lesion and placing the remainder of the gland on surveillance, we expect to observe that a substantial proportion of patients *will* progress and require either additional focal ablation or radical intervention. This might be viewed as the cost of using FT as an extender of AS to preserve quality of life, but patients will have to be prepared to possibly undergo repeated treatments. In the long term, the cost and morbidity of surveillance and repeated FT are not known. Additionally, the morbidity of radical intervention after repeated FT is unknown. The incremental cost-effectiveness of the entire FT strategy remains to be defined.

Third, FT may have to be further optimized using adjuvants to obtain lasting oncologic control. Excellent outcomes with wide excision as compared to mastectomy for breast cancer have often been cited as the model for successful prostate FT. However, although wide excision in breast cancer is routinely paired with adjuvant radiotherapy, there have been no trials exploring adjuvants such as whole-gland irradiation in conjunction with focal ablation in prostate cancer. There are several promising adjuvants in early development, and the downstream challenge is to demonstrate clinical safety and efficacy of such adjuncts in human trials.

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